

Process Patterns for Data Trustee Models

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Authorization Management

Manual Consent

Participating Actors: Data Provider, Data Trustee, Data Consumer

Description: The process begins with the data consumer sending a request for data access to the data trustee via a message. This request is received in the data trustee's process via an incoming message event, which triggers its processing. In a subsequent step, the data trustee forwards the request by obtaining consent from the data providers in a send task. Here, a corresponding message is transmitted to the data provider's process, with the data provider receiving this notification as a start signal for their own process.

In the data provider's process, a detailed review of the request first takes place. In a user task, the responsible person gains insight into the intended uses and checks, among other things, the intended purpose of the requested data. After completing this review, the workflow reaches a decision point, which is implemented via an exclusive gateway. This gateway sets the course for two alternative paths: In the event of a positive decision, consent is sent via an intermediate event, while in the event of a negative decision, an intermediate event is also activated to transmit the rejection.

In the positive case, a "consent" message is transmitted to the data trustee via the intermediate event. This reception occurs in their process via an intermediate catch event specifically designed to receive consent. The receipt confirmation leads to completion in both processes, for the data provider and the data trustee, via defined sequence flows, which are marked by end events as a successful completion ("Data provided").

If the data provider rejects the request, this happens analogously via a separate intermediate event that sends a rejection message. This message is transmitted to the data trustee via the corresponding message exchange, where it is received in a dedicated intermediate event. Subsequently, the rejection is communicated to the data consumer in a send task within the data trustee's process. This flow of information results in end events that document the aborted or denied data access both in the data provider's process ("Data release denied") and at the data trustee ("Data not provided").

The control of the various workflows and decision paths is enabled by the targeted use of gateways. The exclusive gateway in the data provider's process clearly decides between consent and rejection, while an event-based gateway in the data trustee process waits in parallel for both responses, consent and rejection. This constellation ensures that the communication between the participating actors is clearly separated and at the same time linked together.

Overall, through the structured message exchange, the data consumer's initial request is transmitted to the data trustee, then passed on to the data provider via the central role of the data trustee, and finally, the data provider's decision, whether consent or rejection, is communicated back to the data trustee and, in the event of a rejection, directly to the data consumer. This coordinated process ensures transparency and traceability of the entire decision-making process for data release.

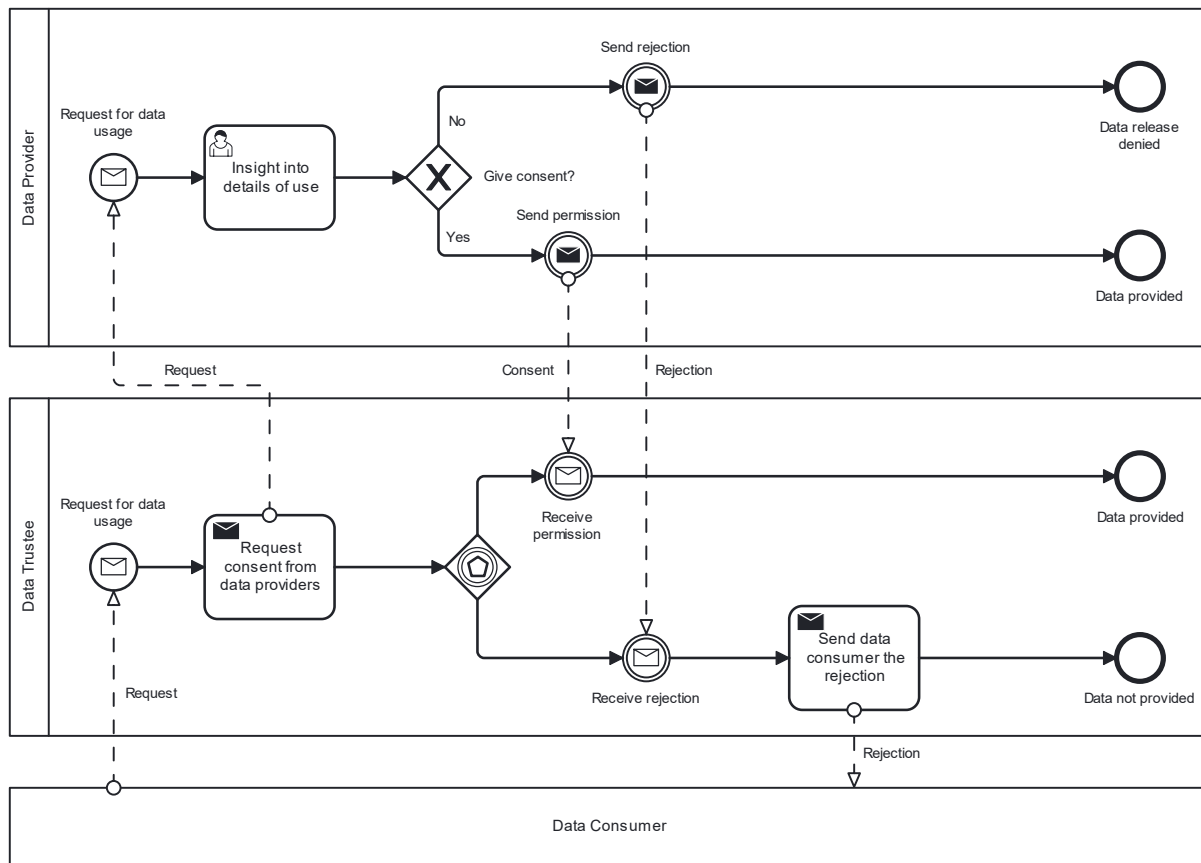


Figure 1. Manual Consent.

Automatic Consent

Participating Actors: Data Trustee, Data Consumer

Description: The illustrated process begins with the receipt of a request to use data, which is triggered by a message from a requesting actor to the central controller. This request activates a message start event that automatically sets the process in motion. Immediately afterward, a system-supported check is carried out, in which compliance with predefined usage rules is evaluated in a service task. Here, an assigned data object flows into the check, providing the necessary information to evaluate the conditions.

As soon as the corresponding service task is completed, an exclusive decision element guides the further course. At this point, it is determined whether the requesting actor meets the specified criteria. If the check yields a positive result, the end of the process is reached via a specific path, signaling that the requested data is released. Otherwise, if the specified conditions were not met, an alternative path is taken. In this case, a rejection message is automatically generated and transmitted to the requesting actor, after which the process ends in a separate end event indicating the non-provision of the data.

The entire process is executed in the context of a collaboration in which, in addition to the central controller, other roles, in particular, a data provider and a data consumer, are integrated. While the data provider, as the rule-setter, implicitly provides the basis for the established terms of use, direct communication between the requesting actor and the central entity takes place through defined message flows. Thus, the initial request is received as a message, and in the event of a rejection, corresponding feedback is provided.

Overall, the BPMN diagram visualizes all essential elements of this process: It depicts the message start, the service task for checking the data conditions, the exclusive gateway for decision-making, the sending task in case of non-compliance, as well as the two different end events documenting the successful completion or the rejection of the data request. This automated decision-making mechanism ensures consistent and efficient processing of requests based on the previously defined rules.

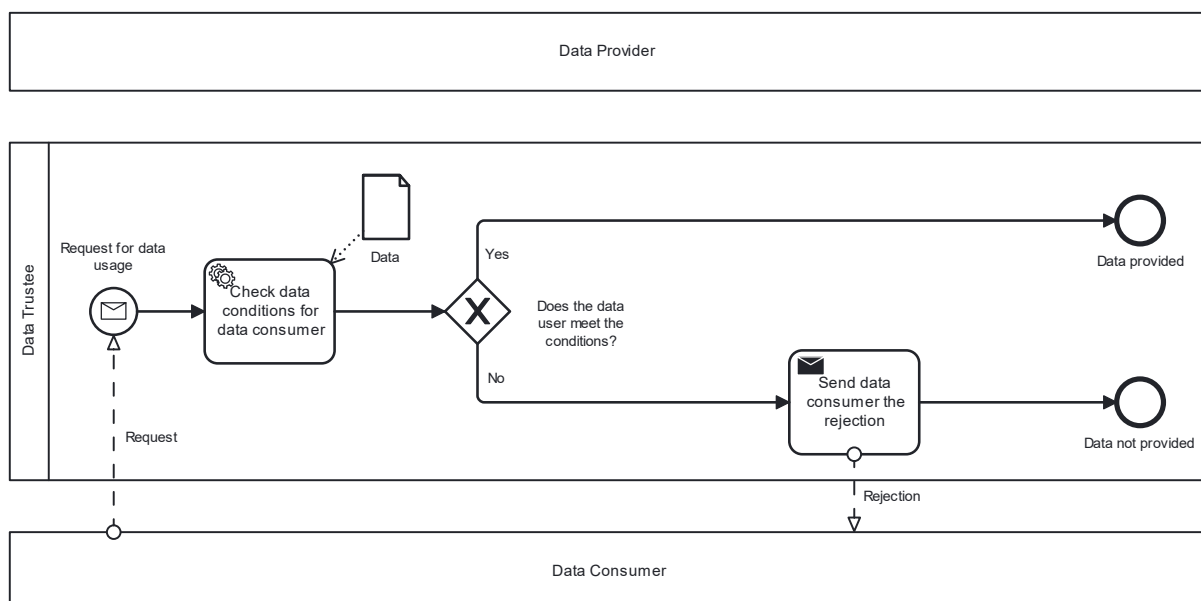


Figure 2. Automated Consent.

Data Access Transparency

Aggregated Data Audit

Participating Actors: Data Provider, Data Trustee

Description: This process establishes a multi-stage communication and processing workflow between the two central actors, where one provides their data for analysis and clear usage evaluation, and the other handles the aggregated analysis. The process starts with the actor indicating that they wish to view the usage of their data. Already at this point, a start event is triggered signaling the request, and this impulse flows into a first user task in which the request is formulated and sent.

Via a defined message flow designated as “Request,” this request is transmitted to the responsible actor in charge of the aggregated data analysis. In the receiving workflow, a process start is initiated by the incoming message event. First, in a subsequent task, all data is selected from a central database. Here, an explicit reference is made to a data store object to ensure that the necessary information is systematically queried.

In the next step, the selected data is aggregated in a separate activity. This aggregation summarizes information such as usage frequencies, general usage patterns, and categories of data usage without disclosing individual accesses or specific transactions. After the data aggregation is complete, the created aggregated report is transmitted back as a message to the requesting actor in a dedicated send task. This return flow, shown in the diagram as a message flow labeled “aggregated Data report,” leads to an intermediate event in the original workflow that waits for the report's receipt.

As soon as this report is received, another user task follows in which the aggregated data is clearly displayed. The final endpoint in this part of the process confirms that the aggregated usage information was successfully displayed. It should be emphasized that both process paths, that of the requesting actor and that of the aggregating actor, are connected to each other via continuous sequence flows so that each activity seamlessly transitions to the next. Communication between the two systems occurs asynchronously via clearly defined message exchange points that control the flow of information. A third actor, designated as a data consumer, appears in the collaborative diagram but plays no active role in the specific workflow.

Overall, this process ensures that, upon the data provider's request, a summarized, anonymized overview of data usage is created, revealing only aggregated key figures and completely excluding individual accesses or specific transactions. The integration of a central data store for selecting the relevant information as well as the clearly defined activities and messaging steps guarantee a transparent and traceable process flow.

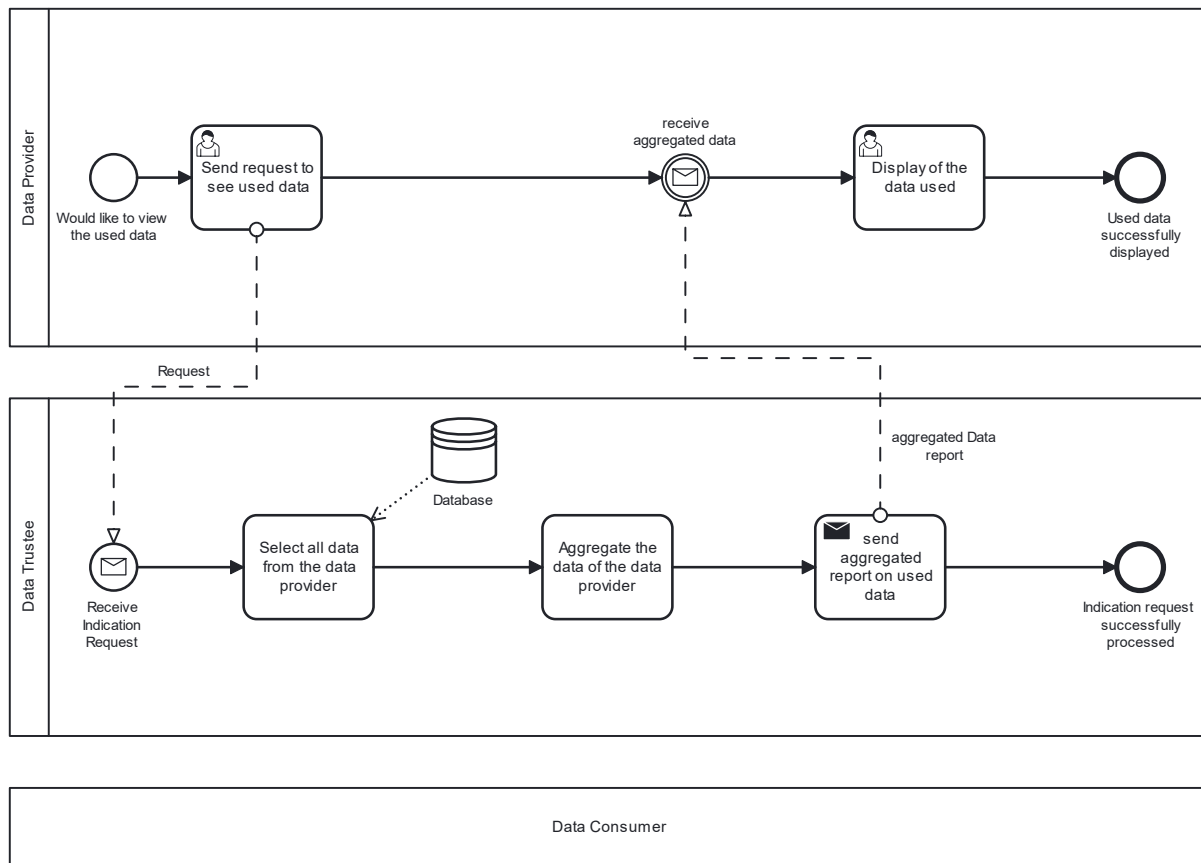


Figure 3. Aggregated Data Audit.

Transparent Data Audit

Participating Actors: Data Provider, Data Trustee

Description: The process starts on the side of the data provider, who signals the desire to gain insight into the use of their data. Already in the initial step, the request to view the used data is triggered using a start event, combined with the initiation of a manual task. Via a defined message connection, this request is transmitted to the data trustee, who thus initiates their review process.

In the data trustee's process, execution begins with the receipt of the request via a message start event. Subsequently, in a first work step, a task is performed in which all relevant data is extracted from a central data store. This task accesses a referenced database to select all data relevant to its use. The raw data obtained is then systematically matched with the corresponding data consumers in a task designed as a loop. This iteration ensures that all connections between the retrieved data elements and the recipients of their usage are comprehensively recorded.

After the assignment has been made, detailed information about the respective usage is prepared in another step, which is also designed as a multi-instance activity. Information such as the participating data consumers, the specific intended purposes, and the times of data use flows into the preparation. After all data aspects have been compiled, a comprehensive report is sent. This is transmitted to the requesting data provider using a send task, a communication realized by an explicitly defined message flow.

In parallel, an intermediate event waits on the data provider's side, coordinated with the receipt of the transparently transmitted data report. After the report is received, a work step follows in which the report is displayed as part of a manual task. If the display is successfully completed, the process on the data provider's side ends with an event that signals the successful completion of the data display.

In summary, the seamless tracking of data usage is guaranteed by close coordination between the two actors, the data trustee, who systematically selects, assigns, prepares, and sends the data, and the data provider, who initiates the request and subsequently receives and views it. The communication flows between the systems and the iterative processing of individual data aspects ensure a transparent and traceable review process.

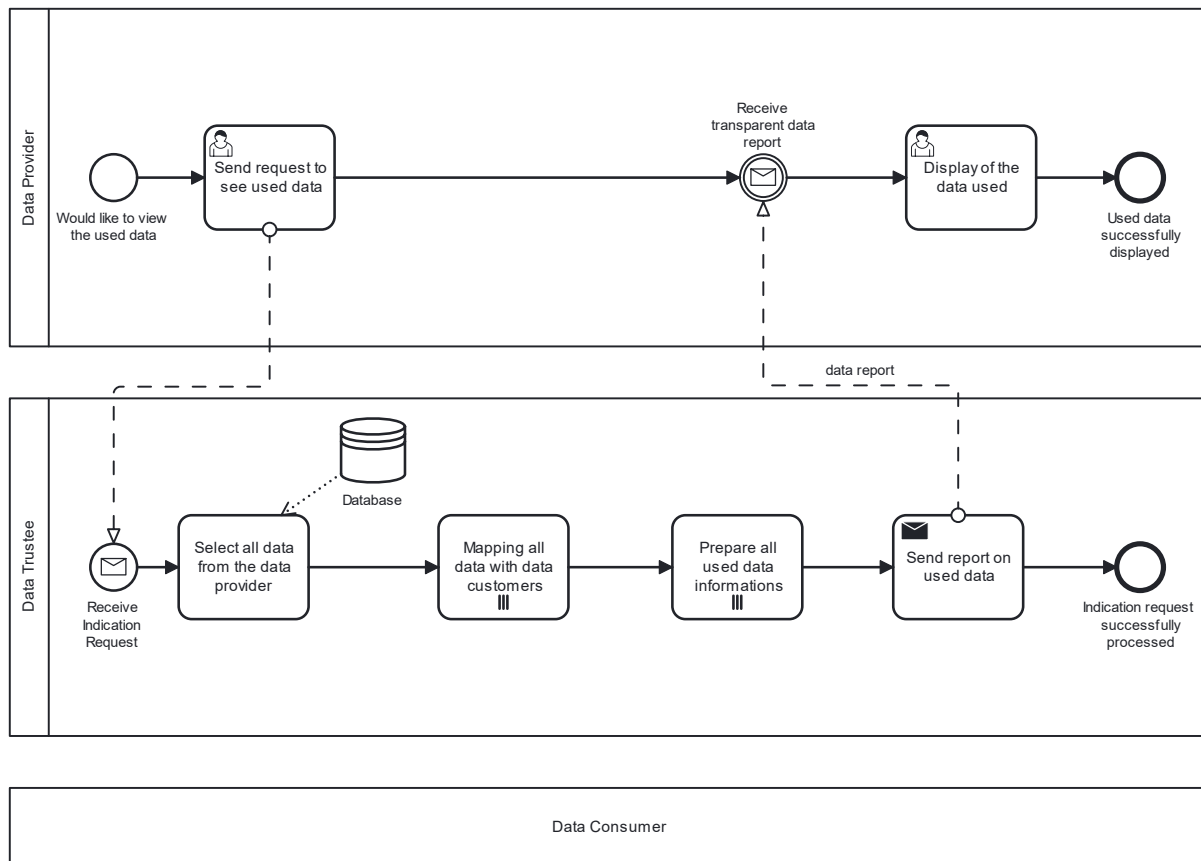


Figure 4. Transparent Data Audit.

Data Disposal

Request-based Data Disposal

Participating Actors: Data Provider, Data Trustee

Description: The presented workflow begins with the actor who wishes to delete the data initiating this request in the form of a start event. A sending process is executed immediately, during which an explicit deletion request is formulated. This request is transmitted directly via a dedicated message flow, designated as “Request”, to the instance of the actor responsible for actually carrying out the deletion process.

On the receiving end, the data trustee activates its process as soon as the corresponding message catch event signals the receipt of the deletion request. This is followed by a systematic identification of the affected data sets. For this purpose, the relevant data inventory is determined in a selection task from a predefined data store, referenced as a database. Once the data to be deleted has been selected, this step leads to another activity in which the actual removal of the identified data is performed. Both processing steps, the selection of the affected data and its subsequent deletion, are linked by a clearly defined sequential flow, ensuring that the information to be deleted is first captured and then immediately removed.

After the successful execution of the deletion operation, the data trustee reaches an end event that confirms the successful completion of the deletion process. This end event is also designed to function as a message source. Via a specified message flow, marked as “Confirmation”, a confirmation is sent back to the original actor who initiated the deletion request. They receive the feedback in a special intermediate event designed to verify the successful execution of the deletion process. Upon receipt of the confirmation, the entire workflow finally ends with a concluding event that signals the successful completion of the process from the perspective of the initiating actor.

The illustration demonstrates how the entire deletion process is structured and comprehensibly mapped through the use of start, intermediate, and end events in conjunction with sending and receiving tasks, as well as clearly defined message flows. The explicit transmission of the deletion request and the subsequent confirmation ensure that communication between the participants, here between the data provider and the data trustee, runs smoothly. The data-specific interactions, namely the selection and deletion of the affected data sets using a referenced data store, are also precisely documented. Overall, this workflow ensures that the deletion request is implemented without delay and that the successful completion of the deletion process is verified through transparent communication.

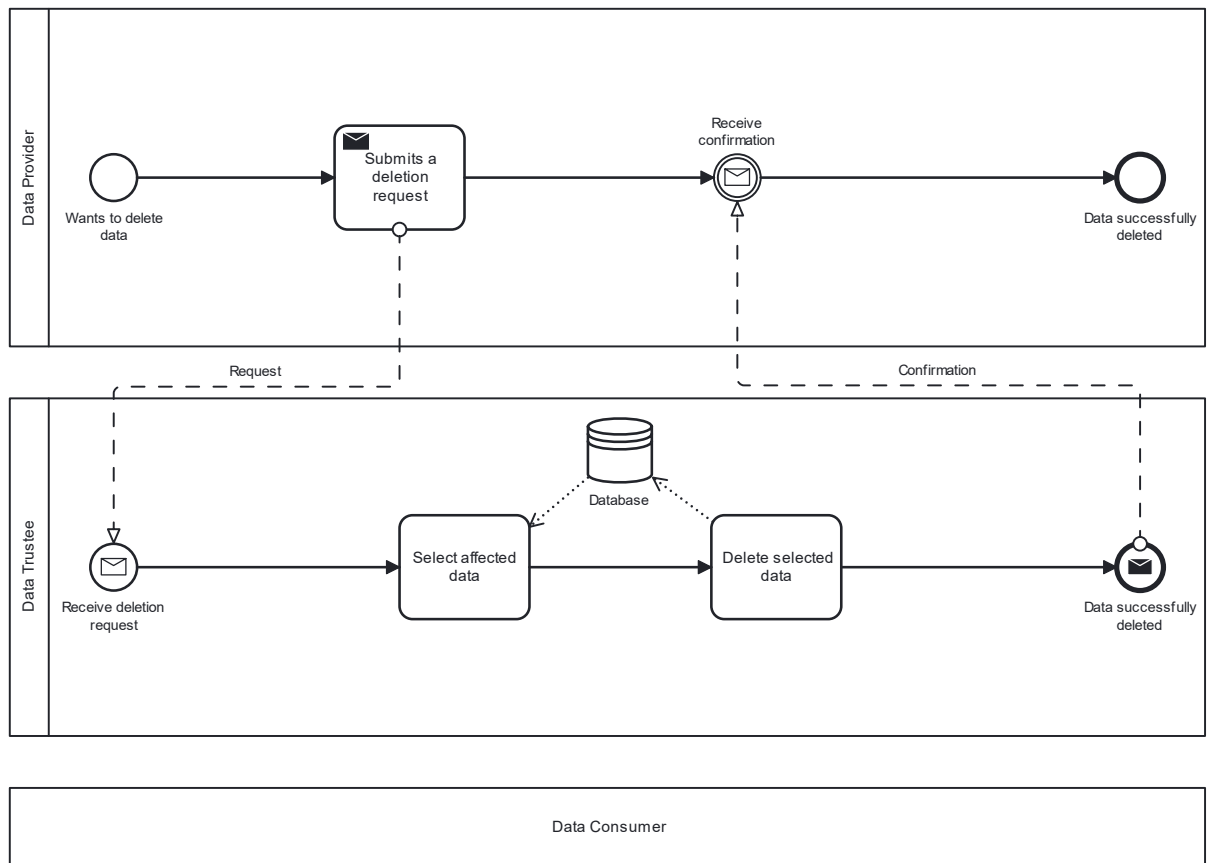


Figure 5. Request-based Data Disposal.

Use-based Data Disposal

Participating Actors: Data Trustee, Data Consumer

Description: In the presented scenario, the deletion of data occurs immediately after the completion of its agreed use through a coordinated message exchange between the involved parties. The workflow is realized by two closely interlinked sub-processes executed by the roles of the data trustee and the data consumer, while the data provider takes on a passive but essential informational role.

Initially, the data consumer signals the completion of data usage through a trigger event for “Data used”. In the course of this, a confirmation is automatically transmitted within its process via a defined send activity. This message flows via a dedicated message exchange to the data trustee, who activates its executable process upon receiving the confirmation message (initiated via a corresponding message start point).

After receiving this message, sequential processing takes place in the data trustee's process. First, the amount of data to be deleted is determined in a service task, the affected data sets are selected, with relevant data being integrated from an external data store, the repository referred to as the “Database”. Subsequently, another automated activity initiates the actual deletion of the selected data sets. During this deletion process, a data output association simultaneously ensures that the procedure is consistently documented in the referenced data inventory.

Upon completion of the deletion operation, an end event marks the successful conclusion of the process at the data trustee. This end event automatically triggers a confirmation message that is sent to the data consumer's process via a defined message exchange. The data consumer receives this confirmation in an intermediate catch event, after which its process is finally terminated with an end event. Thus, the entire deletion process, from initiation to data selection and deletion to the final confirmation, is handled completely and transparently.

In addition to this dynamic information exchange, the data provider is integrated as a separate role within the collaboration. Although no detailed activities are modeled in its own pool, the data provider's participation serves the final transfer of information: After the successful completion of the deletion operation, they are also informed about the process so that all parties are promptly aware of the procedure's conclusion.

The close integration of the two main processes occurs via clearly defined message exchange flows. The first flow connects the send activity in the data consumer's process with the message start point in the data trustee's process, thereby initiating the deletion action. The second message exchange ensures that the successful deletion of the data is reported back by the data trustee and registered by the data consumer as a receipt of confirmation. Furthermore, the use of a central data repository in the selection and deletion step ensures that all data-specific information is managed in a consistent location.

Overall, this workflow realizes an automated, traceable, and auditable data deletion that occurs immediately after the specified intended purpose has been fulfilled and is accompanied by transparent communication among all involved actors.

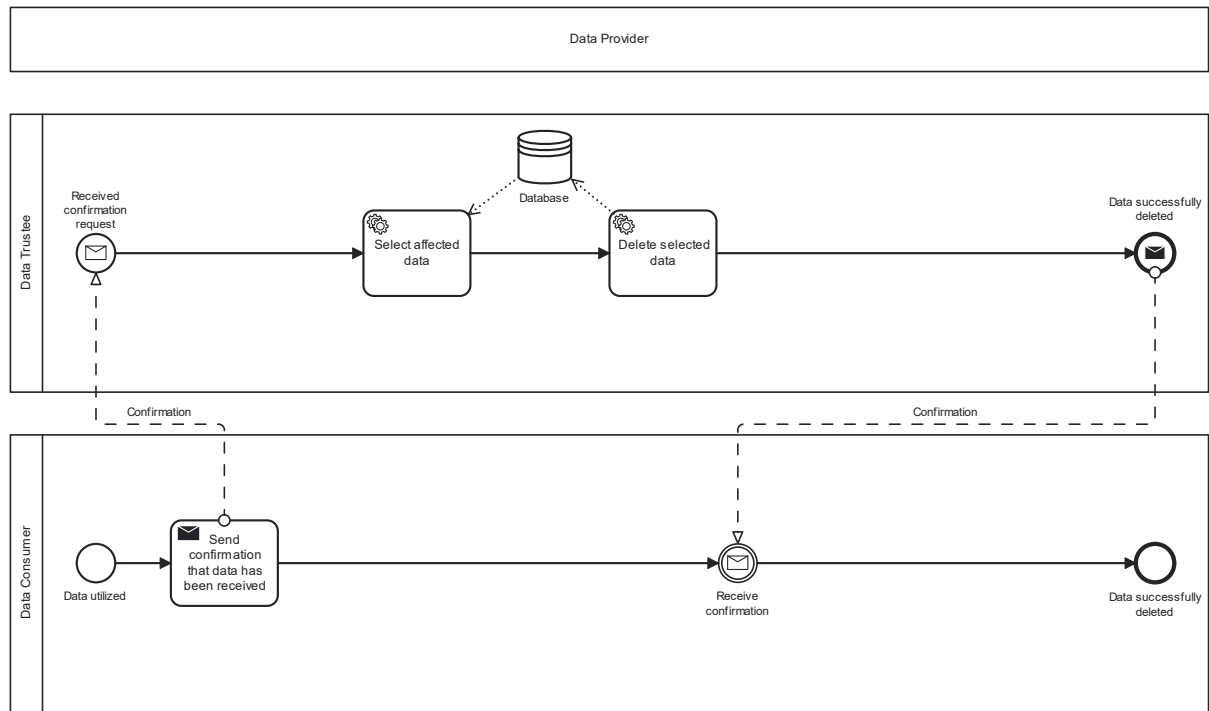


Figure 6. Use-based Data Disposal.

Time-based Data Disposal

Participating Actors: Data Trustee

Description: The workflow starts automatically as soon as the previously determined time is reached. The timer-controlled trigger signals that the retention period has expired and the conditions for final data deletion are met. At this moment, an automated activity is initiated that selects the affected data elements from a persistent store (represented by a database). Access to the data store occurs via a defined data input association, which ensures that all relevant data sets meeting the deletion criteria are identified.

Immediately afterward, the process transitions to another automated service task that permanently removes the previously selected data. This deletion activity is initiated by a direct sequence from the selection task. During this phase, the database is integrated again, this time via a data output association, to ensure that the deletion process is correctly and sustainably recorded in the data store. This guarantees the permanent removal of the relevant data sets.

Once the deletion process is complete, the workflow ends in an end event that signals the successful completion of the operation. Although the graphical representation does not contain an explicit notification activity, the process envisions that the responsible data trustee informs the data provider about the successful removal of the data. This ensures that all involved parties, especially the data provider, who originally defined the retention period, are promptly informed about the completion of the deletion process.

In the overall picture, the process is carried out within a collaborative environment in which, in addition to the data trustee, the data consumer and the data provider are also listed as participating actors. Although the modeled process control focuses primarily on the data trustee's tasks, the other stakeholders are taken into account within the overarching collaboration. Through the use of clearly defined data associations and linear transitions between the individual service-oriented tasks, a precise and automated execution of data deletion is guaranteed as soon as the predefined period has expired.

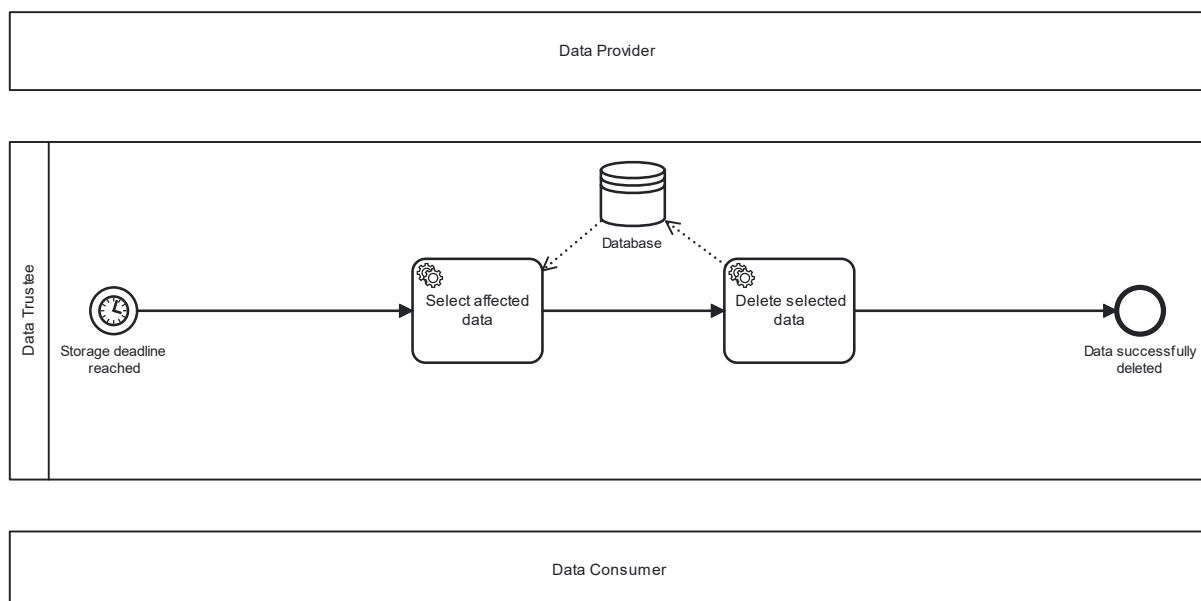


Figure 7. Time-based Data Disposal.

Data Forwarding

Continuous Data Forwarding

Participating Actors: Data Provider, Data Trustee

Description: The workflow starts as soon as the data provider initiates the continuous generation of data. Immediately after this start, the process flows into a first decision unit, where either the regular upload process is continued or, in the event of a deviating event, a rollback of the process is initiated. At this point, the continuous upload process is realized in a recurring sub-process designed as a loop mechanism.

Within this sub-process, the individual cycle begins with a start event to initiate the upload. This is followed by a call activity to encrypt the data, where the information to be secured is taken from a defined data object. The result, the encrypted data, is then passed on in order to publish the data stream in a subsequent service task step. Through this publication, the encrypted payload is transmitted to the recipient via an outgoing message flow.

Parallel to the sending of the data, a synchronous waiting phase also takes place in the same sub-process: An intermediate catch event blocks the workflow until a confirmation of the successful storage process is reported back by the data trustee. This confirmation is received via a message flow, which marks the upload cycle within the sub-process as successfully completed. Additionally, the sub-process is equipped with a distinct boundary event that can activate a deviating path if a cancellation impulse is sent by the data trustee. If such an event triggers, the flow is routed back to the decision unit so that the upload process is terminated.

In the parallel process, which is located at the data trustee, the data stream is first received via a message event. Following this receipt, the data is checked for correct formatting and compliance with regulatory requirements in a first inspection step. Should there be any objections here, an adjacent boundary event catches the error, causing an alternative path to lead into a decision unit. This path ultimately leads to the end event that signals the failure of data storage and sends a cancellation signal back to the data provider.

If, on the other hand, the review process concludes without objections, the workflow is transferred to a preparatory process step, which is mapped in the form of a call activity. This is followed by another call activity responsible for the actual data storage process. Here, too, a specific boundary event is integrated, which takes over the alternative path in the event of a storage error and ultimately leads back to the “Data storage failed” end event. If the storage process is carried out successfully, the process leads to an end event that documents the successful completion. A confirmation is then also sent to the data provider via a corresponding message flow.

Message flows play a central role here: The data provider transmits the encrypted data to the data trustee via the published data stream, while in the event of success, the positive feedback concludes the upload cycle. In the event of an error, the transmitted cancellation signals cause the ongoing data transmission process at the data provider to be interrupted or terminated.

The overall process is characterized by its iterative nature, the upload loop is run through at regular intervals or continuously until the data provider deliberately stops the process. This bidirectional message exchange ensures that successful transmissions as well as any errors or aborts are communicated at an early stage and processed accordingly.

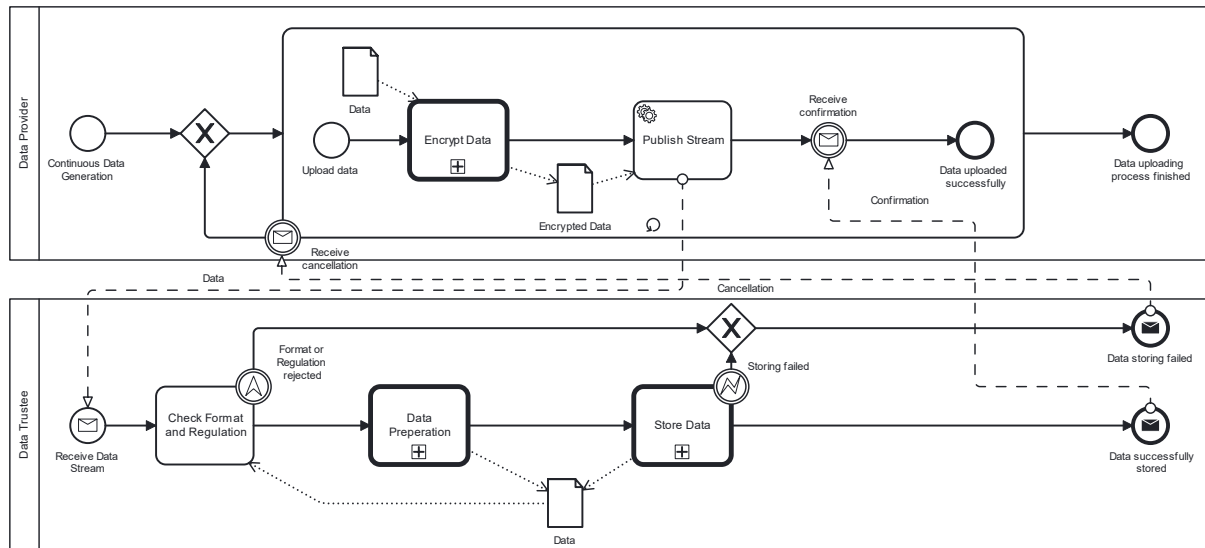


Figure 8. Continuous Data Forwarding.

Periodic Data Forwarding

Participating Actors: Data Provider, Data Trustee

Description: The presented business workflow describes a recurring data transmission cycle in which two systems work together in a coordinated manner. The process is initiated on the data provider's side by starting a periodically triggered upload process. Right at the beginning, it is triggered by a strictly defined upload signal that starts the process either as a new cycle or as the resumption of an interrupted sequence. For this purpose, the trigger first flows into a decision element which, whether from the initial start signal or from a feedback loop as a result of an error event, transfers the process to the next step.

After this decision, a time-controlled intermediate event follows, representing the predefined time interval. Only after the timer has expired is a call activity started to encrypt the pending data sets. In this step, the data to be transmitted is prepared by being encrypted via a separate call and then integrated into a corresponding data object. The output of this encryption is immediately passed to another technical activity whose task is to publish the encrypted data stream.

As soon as the data has been published, an event-based gateway is used, which triggers the next step by receiving feedback. Essentially, two alternative events are caught here: On the one hand, a confirmation message can arrive indicating the successful upload and concluding the cycle. On the other hand, if a cancellation message arrives, triggered by errors in the downstream processing operation, the process is not marked as complete but returns via a feedback path to the decision element so that the upload can take place again in the next cycle.

In parallel, an independent but content-synchronized sub-process runs on the side of the data trustee. This begins with the receipt of the data stream sent via the message channel. Immediately afterward, the data is checked in a first processing step to see whether it complies with the required formatting specifications and whether the necessary consent has been given. This check is secured by a boundary event attached to the activity, which immediately intercepts the flow in the event of non-compliance or missing consent.

If the check is successful, the next step is a data preparation call activity in which the received data sets are further structured or transformed. This is followed by another call activity in which the prepared data is stored in a data repository. An error-proofing mechanism is also implemented for storage on the data trustee's side: An error event linked to the storage intervenes if problems arise during the filing process. Both in the case of a failed format or consent check and in the case of a storage process ending in error, the process paths converge via an exclusive gateway and are terminated via an end event that generates a cancellation message as a consequence.

If, however, data preparation and storage occur without error, the process reaches an end event that sends a confirmation back to the data provider via a message. This receipt of the confirmation, mediated by the event-based gateway in the originating system, then leads to the successful completion of the current upload cycle.

Communication between the participating systems takes place via clearly defined message flows: The encrypted data stream is transmitted directly from the data provider to the data trustee, while the feedback, confirmation or cancellation, controls the further course of the periodic upload. In addition, the exchange of data objects is explicitly taken into account in the model through corresponding data input and output associations so that the results of the preprocessing steps are passed on consistently and comprehensibly.

Overall, the use of timers, call activities, and event-based as well as exclusive gateways creates a robust loop for periodic data transmission that detects error conditions early on, communicates them across processes, and thereby enables repeated attempts at transmission.

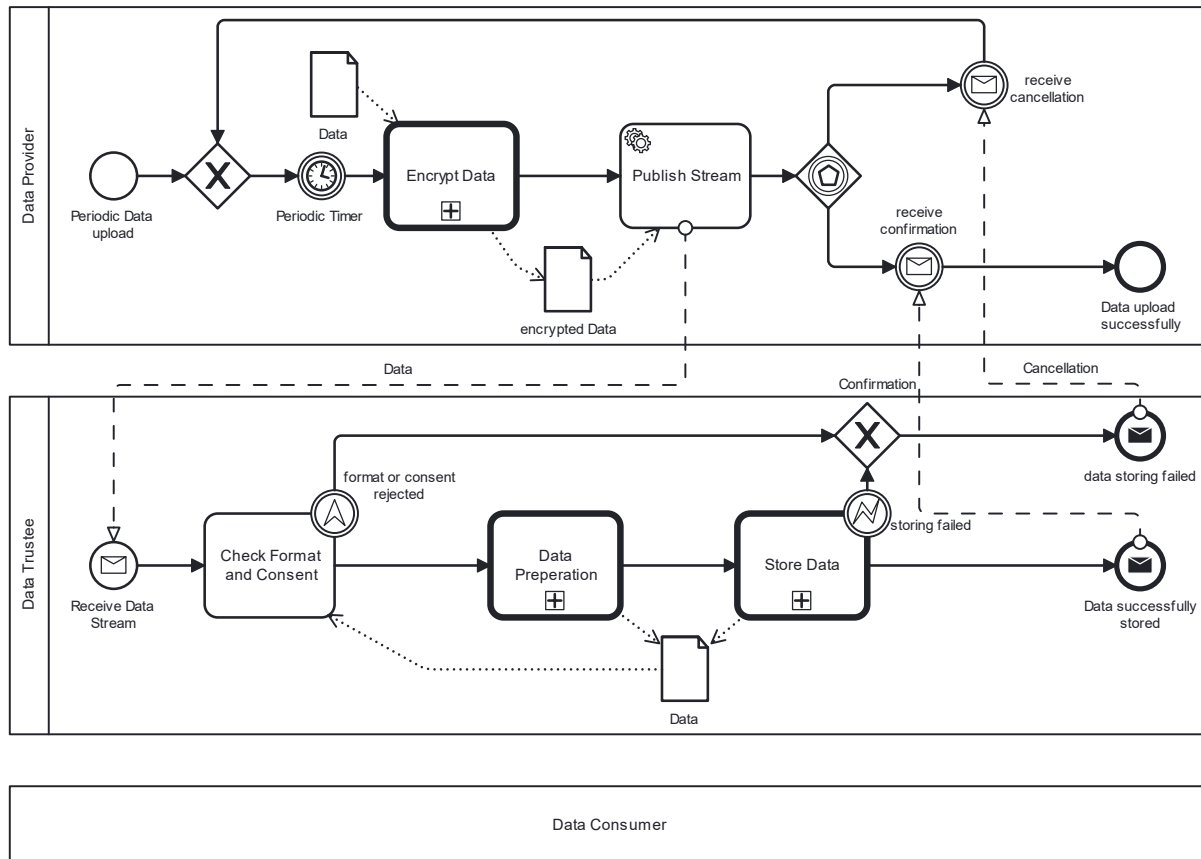


Figure 9. Periodic Data Forwarding.

Single Data Forwarding

Participating Actors: Data Provider, Data Trustee

Description: The illustrated workflow provides that a one-time transmission process is first started on the side of the data provider. In a first step, the data provider decides to transmit the prepared amount of data. Then, an automated activity is triggered in which the content to be sent is encrypted. Here, the originally existing data object is converted into an encrypted state so that its confidentiality is guaranteed during the transfer. The content transformed in this way is then released in a manual task to carry out the upload, with the data provider transmitting the encrypted data to the recipient via a secured connection.

Once the upload is complete, the data provider awaits feedback from the data trustee. At this point, the process branches into two parallel paths using an event-based gateway: On the one hand, it waits for the receipt of a confirmation; on the other hand, there is the possibility that an abort message is received. In the event of success, the receipt of the confirmation leads to an end event that signals the successful transfer. If, however, an abort message is registered, the process ends with a corresponding error message.

In parallel, a separate workflow was designed on the data trustee's side, which begins its processing with the receipt of the transmitted data. The data trustee activates its process by receiving a message containing the content to be transmitted. In a first processing step, the data is checked regarding its format and compliance with the required consents. If this validation is successful, data preparation is carried out in a further automated call, a measure necessary for the subsequent storage in the data trustee's system.

Following this, an attempt is made in a separate activity to transfer the prepared data into the system. This storage process is continuously monitored: Should problems arise here, or should deficiencies in the data format or in the granted consent already be identified in the previous step, error events (boundary events) coupled to the respective activities take effect. Both error cases lead via a decision node to an end event signaling the failed storage process. From here, an abort message is sent back to the data provider via a message flow.

If, conversely, the storage process concludes correctly, a confirmation message is generated in the end event of the successful data storage by the data trustee. This is sent to the data provider via a defined message exchange, who then records the positive conclusion of the transmission process.

Communication between the two parties occurs via dedicated message flows: First, the upload of the encrypted data sends a message to the data trustee, while later either a confirmation or an abort message flows back. This exchange is supplemented by explicitly modeled data associations, which ensure that relevant data objects, for example, the original and the encrypted data package, are consistently available during the various processing steps.

Overall, the workflow represents a clearly structured, one-time data transmission in which both automated and manual activities are integrated. The parallel operation of both processes with synchronous message exchanges ensures that the data provider is always informed about the success or failure of the data transmission. In addition to the regular activities, the modeling includes specific events, such as the event-based gateway, the intermediate waiting or error events, and the error-handling mechanisms bound to activities, all working together to ensure a secure and traceable transmission process.

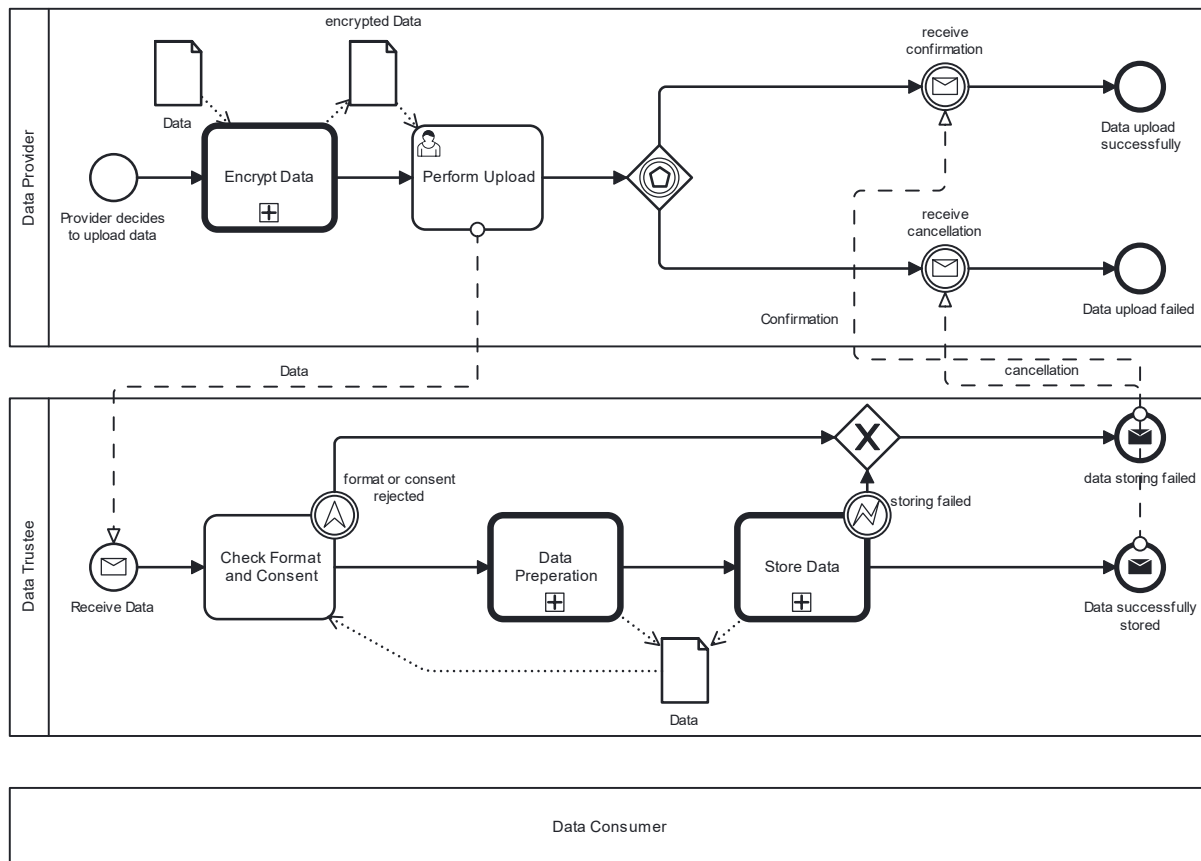


Figure 10. Single Data Forwarding.

Meta Data Forwarding

Participating Actors: Data Provider, Data Trustee

Description: The business workflow begins on the data provider's side by first recording references to the locally available data. This initial step marks the starting point of the process chain, in which no physical data sets are transferred; instead, only their associated meta-information is prepared. Subsequently, in an automated service step, the existing data material is prepared so that the relevant metadata can be extracted and transferred into a corresponding data object.

In the following manual step, the conditions for the future use of the data are checked or determined. Here, a user task ensures that the framework conditions and requirements for the metadata are correctly and completely documented. Building on this preparatory work, the meta-information is actually created in a further automated step. The generated result is stored in a defined data object, which serves as the basis for further communication.

After the necessary information has been compiled, a request to create a data reference is transmitted to the data trustee via a send task. The sending of this message triggers the start of a second process on the recipient side, at the data trustee, which serves for the further review and storage of the metadata.

The data trustee receives the request via a dedicated start event and, in a subsequent service task, takes over the review of the format and the regulatory requirements of the transmitted meta-information. This step ensures that the data meets the necessary quality and compliance standards. Should a deficiency become apparent here, for example, if the format does not meet the specifications or regulatory requirements are not met, this is signaled by a boundary event tied to the service task. Analogously, an error can also occur during the subsequent storage process in a call activity, which is caught via an alternative boundary event.

In the event of success, the “Store Data” step is executed within the scope of the call activity, effectively storing the meta-information, as a representative reference for the locally available data. The successful completion of this process triggers an end event, which signals to the data provider via a message flow that the storage process was carried out successfully. In the event of an error, however, an alternative end event is activated, which sends a refusal message to the data provider via a corresponding message flow.

On the data provider's side, after the request has been sent, the process waits at an event-based gateway for feedback. Two parallel reaction options are available here. If a success message is received, this leads via a clearly defined flow to an end event marking the successful completion of the entire process. If, on the other hand, a rejection message is received, it is first caught via an intermediate loop. An exclusive decision node then checks whether it makes sense to run the process again. If the decision is made to try again, the workflow is routed back to an earlier point so that the data can be prepared again and the request step repeated. If, conversely, a repetition is waived, the process ends with a negative result, signaling the cancellation of the forwarding.

Overall, this workflow allows exclusively metadata to be exchanged between the participating actors. The actual data sets remain with the data provider, while the validation and storage of the meta-information ensure what data is available at the original owner. The close coordination between the automated service tasks, manual review steps, flexible decision points, and defined message flows guarantees a secure, traceable, and repeatable, robust process flow.

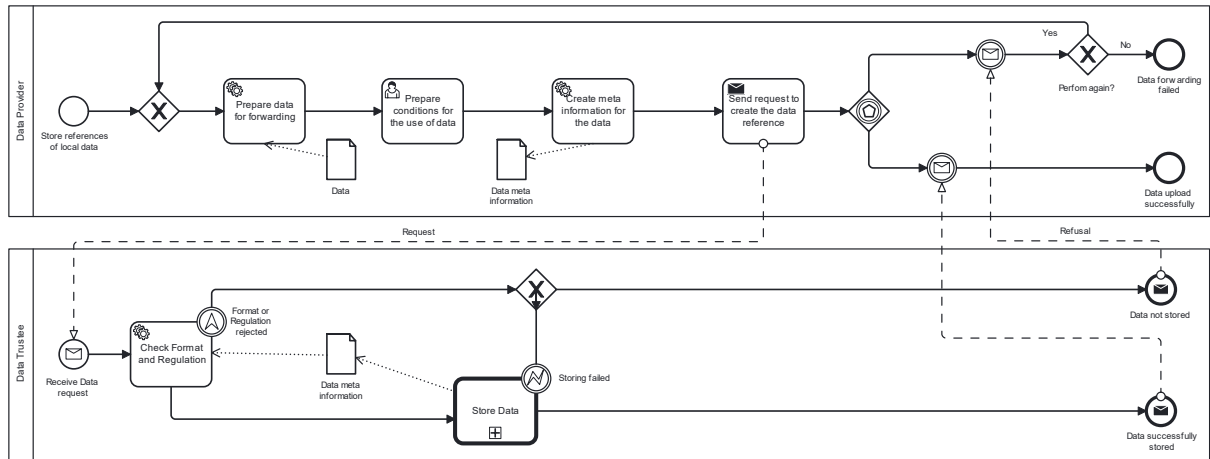


Figure 11. Meta Data Forwarding.

Data Preparation

Data Standardization

Participating Actors: Data Trustee

Description: The workflow starts with the receipt of the data sets, which immediately become the focus of a structured review process. In the first phase, the responsible data trustee ensures that all incoming data complies with the established norms and standards regarding structure and terminology. A detailed review takes place to systematically determine whether the available data sets are complete and compliant with the established requirements.

If it is discovered during this review that individual data or data sets deviate from the defined specifications, the data trustee falls back on predefined conversion rules. These rules make it possible to purposefully transform the identified discrepancies so that the data achieves the desired uniform state. The use of these rules guarantees that a consistent data basis is created even if there are initial deviations.

The corresponding BPMN diagram visualizes this sequential workflow and, through clear activities and decision points, illustrates how data input first occurs, followed by the conformity check, and, if necessary, an alternative path is taken to apply the conversion rules. This structured representation clearly shows that after successful transformation, the entire process ends in a uniform and standardized database.

Through this methodically precise workflow, the data is prepared in such a way that it is not only easier to process further but can also be used reliably and consistently in other contexts. The combination of norm-based checking and targeted adaptation thus ensures high data quality and supports smooth integration into downstream processes and applications.

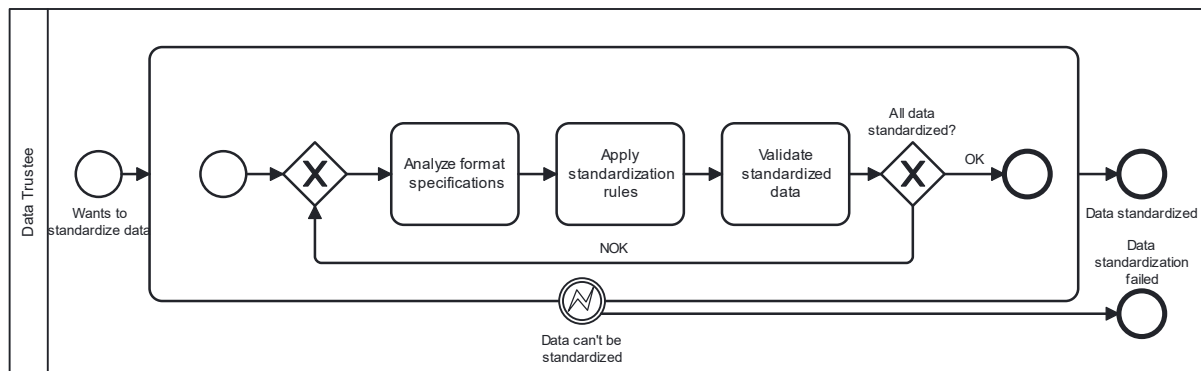


Figure 12. Data Standardization.

Data Transformation

Participating Actors: Data Trustee

Description: In this workflow, raw data is first received and subjected to a thorough review. The responsible data trustee systematically checks whether all required data fields are present, whether the existing format complies with specifications, and whether inconsistencies occur in the data sets. Any discrepancies, such as missing information or unexpected format deviations, are immediately identified and cleaned up in a first step to ensure data quality.

In the following step, the actual conversion of the validated data takes place using firmly defined rules. This transformation step includes, for example, the merging of several source tables as well as the adaptation of value ranges to consolidate heterogeneous information into a uniform, useful format. During this process, all changes are logged without gaps so that it can be traced at any time which adjustments were made and as what decision-making basis they served.

The structured BPMN diagram maps the logical sequence of the individual activities, starting with the initial quality check, continuing through the specific rule-based conversion, and ending with the final logging. Through the use of decision elements and defined transitions, it is ensured that only verified and fully transformed data sets are available for downstream applications. The documented process sequence thus enables transparent, traceable, and reproducible data preparation.

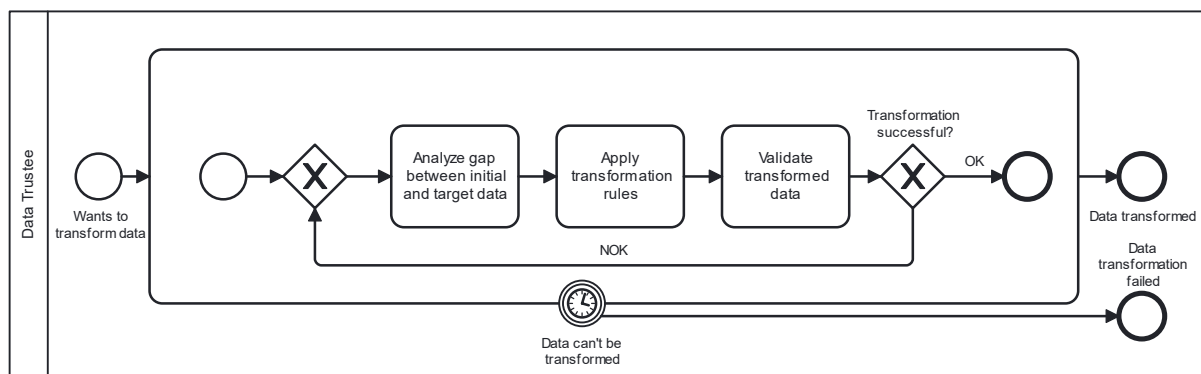


Figure 13. Data Transformation.

Data Retrieval

Single Data Retrieval

Participating Actors: Data Trustee, Data Consumer

Description: The described workflow starts as soon as a data request is triggered by the requesting actor. First, a start event ("Data Request") is activated in the corresponding process area of the data consumer, which transitions into a user task where the specific application for data provision is formulated. Via a defined message flow ("Request"), this request is forwarded to the data trustee's area, where it is received at a start event ("Receive Data Stream").

In the further course, the data trustee takes over the review of the request. The incoming message leads directly into a sub-process ("Authorization Management"), which is implemented as a call activity. This step takes on the central tasks of the authorization check; under certain circumstances, it can verify whether the consent of an additional actor, exemplarily referred to as a data provider, is also required. The result of this check is verified by an exclusive gateway ("Allowed?"). At this decision point, the workflow branches into two paths.

If the result is evaluated positively, the "OK" path comes into effect. Subsequently, a service task ("Prepare Data for Download") is started in which data preparation is realized. Within this task, a data input association is used to access a previously defined database from which the required information is extracted. The result of the preparation is stored in a data object via a data output association and finally passed on via a sequence flow to an end event ("Data successfully prepared"). This end event triggers another, separate message flow through which the prepared data is transmitted as a message ("Data") to the data consumer.

If, on the other hand, the decision gateway selects the negative path ("NOK"), the procedure is terminated in an end event ("Data preparation failed"). This end event is linked to a corresponding message flow that signals to the requesting data consumer that data preparation has failed ("Reject").

On the data consumer's side, communication is supplemented by an event-based gateway that selectively activates two possible intermediate events. If the data consumer receives the positive feedback via the message flow (caught at an intermediate event "Receive data"), this leads via a final sequence flow to an end event that marks the successful completion of data consumption ("Data successfully consumed"). In the case of the negative message flow, which is received via an intermediate event ("Receive message about fail"), the process ends in an error case via its own end event ("Data consume failed").

The modeled collaboration encompasses three actors: the executing data trustee, the initiating data consumer, and an additionally referenced data provider who may be informed as part of the authorization check. The integration of data objects and data stores also enables a clear assignment of the used information: During the service task, the necessary input is obtained via a data store (represented as a database), which is then summarized in a separate data object.

Overall, the coordination of message flows, activities, gateways, and data associations ensures that the request is clearly communicated right at the beginning, that all necessary checks and data preparations are subsequently carried out, and that the final state, be it the successful transmission of the data or the cancellation in the event of an error, is clearly documented and communicated. The structured workflow thus ensures targeted, one-time data provision via a secure connection, which is considered completely finished once the transmission is complete.

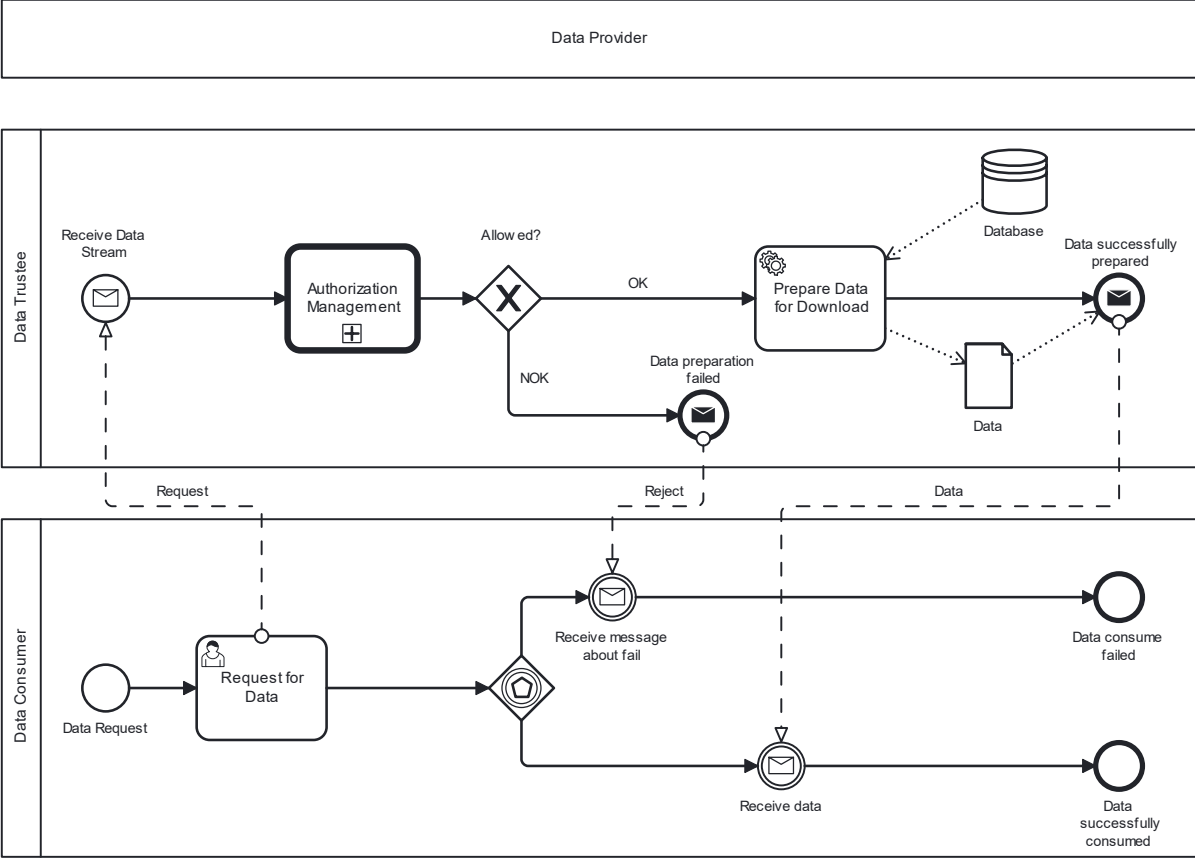


Figure 14. Single Data Retrieval.

Continuous Data Retrieval

Participating Actors: Data Trustee, Data Consumer

Description: The illustrated workflow regulates the continuous provision of data between the involved roles. Initially, the process is initiated in the data trustee's area of responsibility by the receipt of a message signaling the arrival of a data stream. Directly after this start, a dedicated sub-process for authorization management is called up, which checks whether the requesting user is authorized to receive the corresponding data. The results of this check flow into a decision modeled in an exclusive gateway.

If the workflow decides in favor of a release, an activity is subsequently executed to check the data not intended for disclosure. This check verifies both data quality and possible inconsistencies, with the results being documented in an underlying data store. This is followed in a further processing step by the concrete preparation of the data for download. Here, a data object is created based on the information stored in the database, which is released as a package for download. If this path concludes successfully, the process in the data trustee's area ends its execution path by signaling the successful preparation of the data via an end event, simultaneously, a corresponding message is transmitted.

In parallel, a separate workflow starts in the data consumer's field of responsibility, which continuously retrieves data. Right upon entering this process, a repeating sub-process designed as a loop is activated. Within this cycle, the process begins with a start event that initiates the downloading of the data. A subsequent service task then sends a concrete request to the data trustee via a message. Once the request has been processed, the process waits via an intermediate event for the prepared data to be transmitted. The receipt of the data occurs via a message that exactly corresponds to the provided data package and signals the successful completion of a download cycle.

In addition to this loop, a boundary event is assigned to the sub-process, specifically reacting to the receipt of an error message. If the data consumer receives a corresponding message indicating, for example, a failed authorization or problems with data preparation, the ongoing cycle is interrupted. This alternative path leads to an explicit end event reflecting the failure of data consumption.

Communication between the two actors takes place via clearly defined message exchange processes. Thus, an initial data request from the data consumer is transmitted as a "Request" to the data trustee, while in the event of success, the data trustee sends prepared information back as "Data". In the event of a rejection result, however, a "Reject" message is sent, triggering the alternative path in the data consumer's process.

Through the combination of authorization management, data preparation, and recurring request loops, a system is realized that enables permanent and monitored data exchange. Integrated quality assurance and the ability to interrupt the process via corresponding messages contribute to ensuring that only valid and released information is continuously delivered to the data consumer.

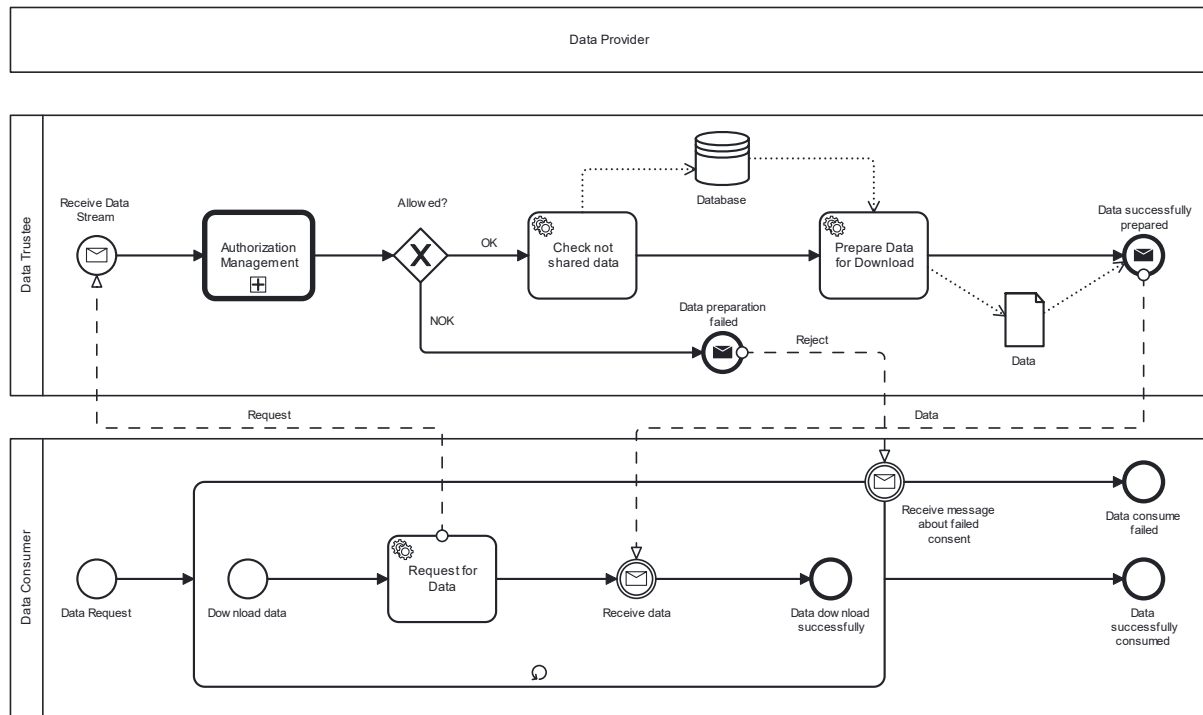


Figure 15. Continuous Data Retrieval.

Local Data Retrieval

Participating Actors: Data Provider, Data Trustee, Data Consumer

Description: In the illustrated workflow, an external actor initiates the process by making a request to access locally stored data. The request is first forwarded to the central instance responsible for the authorization check and the coordination of the participants. After this request arrives, an initial step for managing access permissions takes place, in which a specialized activity for authorization checking is started. Here, it is checked whether the necessary consent criteria are met. The result of this check is evaluated at a decision point, where there is the option to either initiate the continuation of the process or directly reject the request.

If the request is evaluated as compliant or approvable, a further work step involves selecting suitable local data providers using metadata stored in an associated database. Building on this selection process, the central instance uses a multi-instance send task to transmit a request for data access to each selected data provider. Here, if several potential providers are considered, multiple requests can be initiated in parallel.

On the data provider's side, the process begins with the receipt of the access request. A catch event reacts here to the incoming message, whereupon a user-driven activity is started in which the requested data set and the intended purpose of use are presented and reviewed. Based on the results of this review, a downstream exclusive decision point decides whether access should be approved or denied. In the event of a denial, an end event is triggered immediately, sending corresponding feedback to the central instance.

If, conversely, access is approved, a defined call for data encryption takes place. In this step, the data set is encrypted within the local infrastructure so that it can be provided, but the physical data remains securely protected against unauthorized access within the provider's environment. The encrypted data set is then communicated to the central instance via an end event.

On the central instance's side, after sending the requests to the data providers, an event-based gateway is activated, steering different paths depending on the incoming response. If the central instance receives a message with the encrypted data set, a specialized call activity for further data preparation is initiated. The preparation may include additional transformation or checking processes, for example, to further secure the data for subsequent use. After successful preparation, the data set is handed over again in encrypted form via a final end event.

If, however, a rejection is reported back by the providers, be it due to unmet consent criteria or an explicit refusal, a corresponding rejection is transmitted via alternative branching paths. The central instance handles the return of the rejection messages, which leads to a conclusion of the process signaling the failure of data provision.

On the side of the original requester who wanted to request the data, a separate process also starts. This begins with a user-driven step in which the request is generated and then forwards the request via its own work step to the central instance. In the further course, the requester waits at an event-based decision point for one of the possible responses. They can either record the successful transmission and subsequent localization of the prepared data set or receive one of the rejection messages pointing to a failed establishment of the access process. In the case of a successful retrieval, a corresponding catch event concludes the workflow and confirms that the data set was transmitted according to the agreed security settings. In the case of a rejection, the workflow ends with an event signaling the failure of data consumption.

The coordination and information exchange among the three actors, the data provider, the central instance, and the data requester, take place via defined message flows running in multiple directions. For example, the initial request is forwarded from the requester via the central instance to the provider, while positive and negative feedback also flows back via this defined communication infrastructure. In addition, explicit call activities, such as those for data encryption and data preparation, ensure that all necessary processing steps occur within the secure environments of the respective participants. Thus, the entire workflow ensures that data is prepared and transmitted exclusively in controlled, authorized, and completely logged processes without leaving the provider's secure local environment.

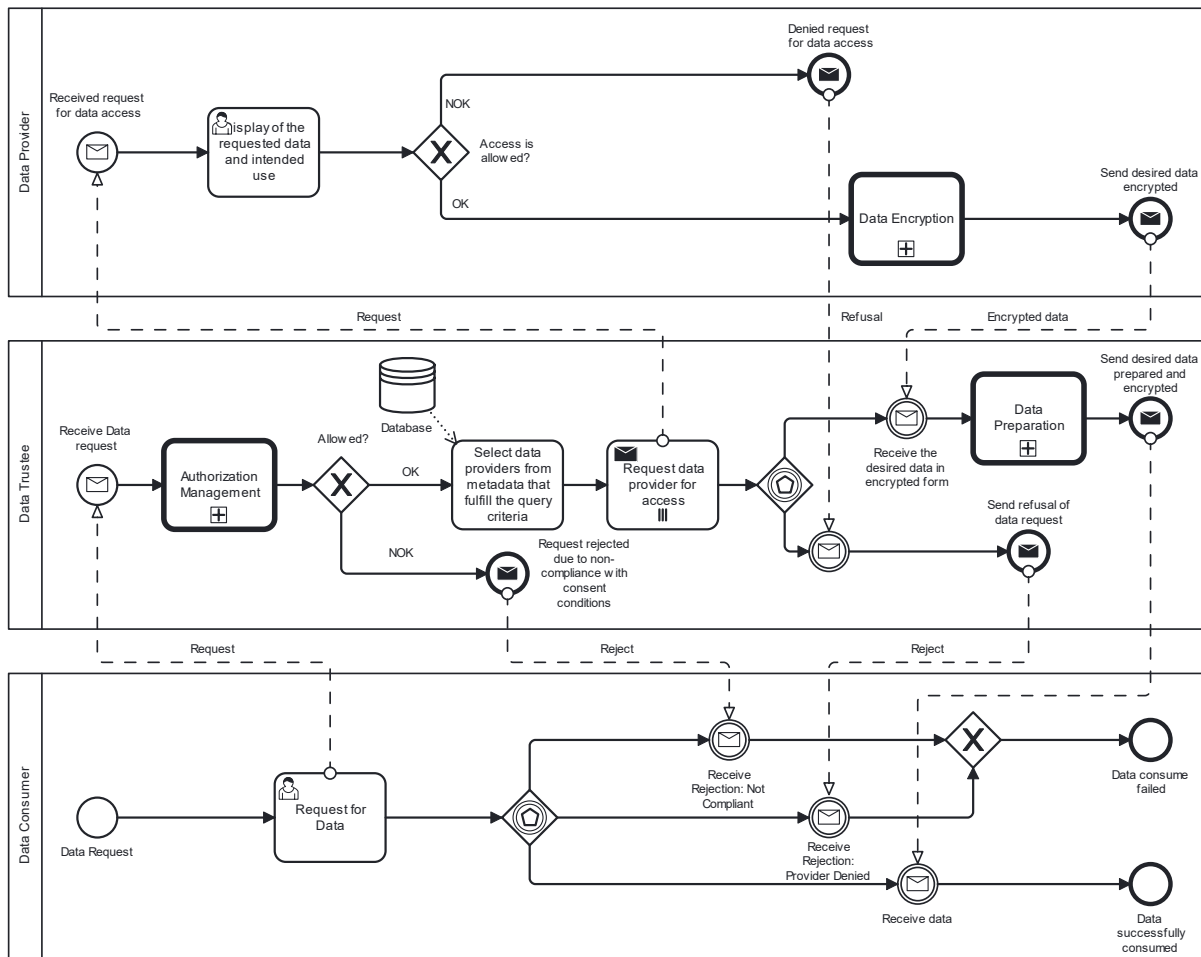


Figure 16. Local Data Retrieval.

Periodic Data Retrieval

Participating Actors: Data Trustee, Data Consumer

Description: The workflow begins with the data provider signaling the desire to subject data to a protective measure, which is formulated in the start event as an intention to encrypt. Immediately after this starting point, an exclusive decision element checks whether the data sets to be processed contain information that should not be transmitted. At this point, it is evaluated whether sensitive, possibly unique data is present that could pose an increased risk if passed on unprotected.

If it turns out that such undesirable information is present, the alternative path is taken, in which the task “Remove unique information” is purposefully executed in a work step. This step serves to convert the data into a form that meets the security requirements placed on it by removing the specific, potentially sensitive details. A data object association is also made here, documenting that the cleaned data is consistently processed further.

If, on the other hand, the decision shows no need for such removal, the alternative path is chosen, which skips this additional processing step. Both paths, the one in which the unique information was removed and the direct route without additional adjustment, subsequently converge in a second exclusive gateway. This element bundles the different process streams and forwards the uniform workflow to the next step.

Finally, the merged path leads into an end event signaling the successful completion of the process. At this moment, it is confirmed that the data is available in the required, protected form, for example, through at least the sanitization of sensitive content, and has thus been made available for further use.

Overall, the process represents a structured workflow in which the desire to modify data is initially triggered, followed by a critical review of the content, a needs-based adaptation of the data, and finally the merging of all paths into a clear end event that documents the successful implementation of the measures.

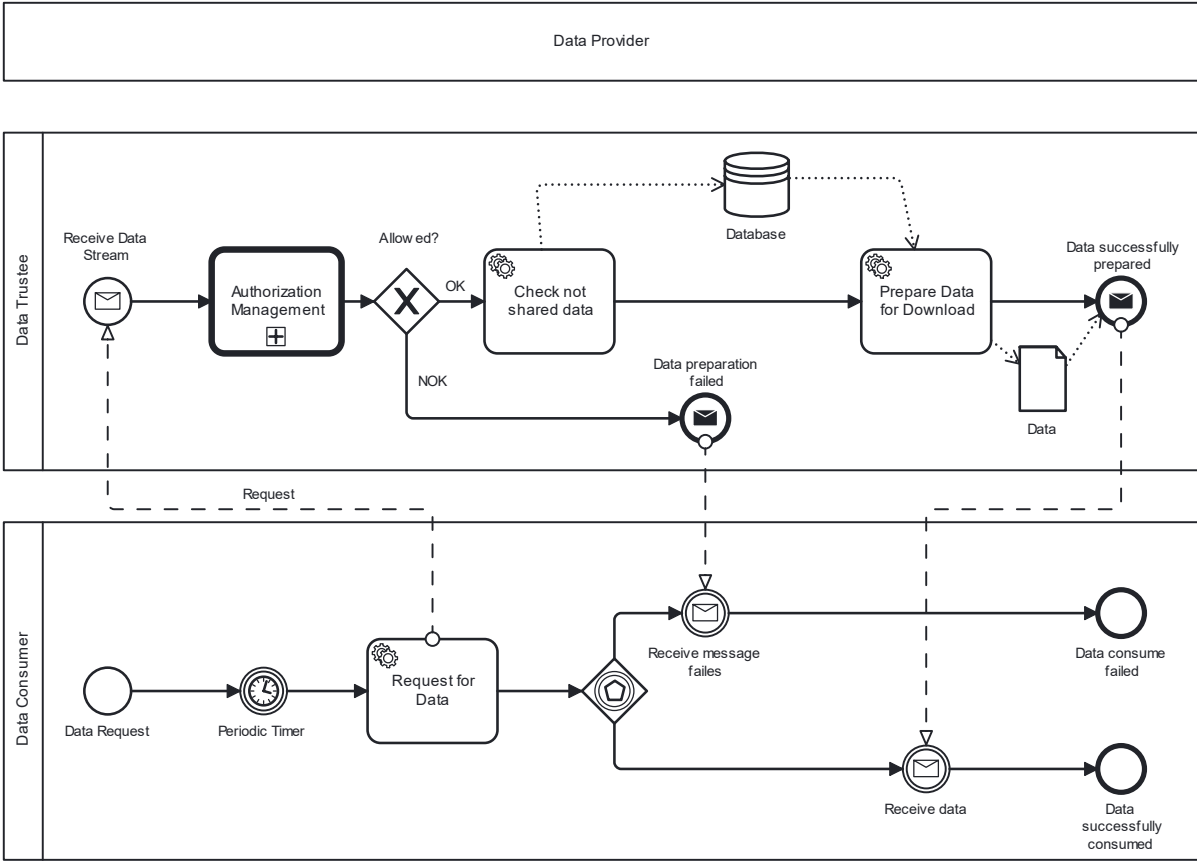


Figure 17. Periodic Data Retrieval.

Data Security

Pseudonymized Data Sharing

Participating Actors: Data Provider

Description: The process is initiated by the data provider, who expresses the desire to transfer the existing data sets into a protected state. This starting point leads into a decision node where it is checked whether the original data contains characteristics that are fundamentally inaccessible to encryption or problematic. If such identifiable elements exist, targeted cleaning is undertaken in a subsequent activity, during which the unique, unprotectable information is removed. Conversely, if the data set has no such elements, the transition occurs without this step.

Regardless of the path chosen, the workflow flows into another merge node that controls the consolidated data flow. Subsequently, a pseudo-key is generated in a special activity. This key is stored in a separate database and serves as the central element for the subsequent transformation step.

In the following activity, the actual conversion of the data takes place. Here, the input data, which may have already been cleaned, is processed together with the generated pseudo-key to create a pseudonymized data set. Ensuring the correct transfer of the relevant information is guaranteed by defined data input and output associations, resulting in the transformed data set as a new, pseudonymized data object.

The process finally concludes with an end event that signals the completion of the encryption or pseudonymization measures. Throughout the workflow, the use of decision nodes and merging gateways provides structured control of the data flow, while the integration of data objects and a key database ensures a consistent and traceable processing of the sensitive information.

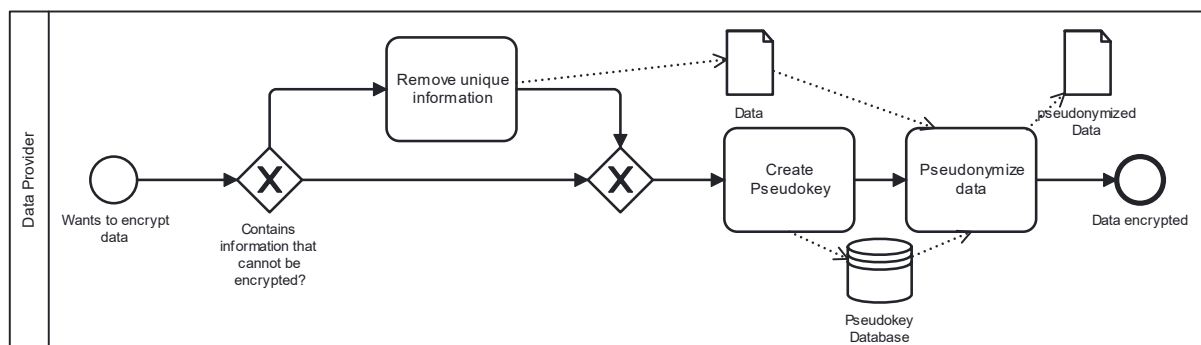


Figure 18. Pseudonymized Data Sharing.

Anonymized Data Sharing

Participating Actors: Data Provider

Description: The workflow begins with the data provider's decision to initiate the procedure to secure the information. Already at this starting point, the trigger signals that it is necessary to edit the data, a step anchored in the intention to encrypt or anonymize the data. Immediately after the start, it is checked whether the transmitted data contains characteristics that preclude direct encryption. A distinction is made here based on a decision unit: If such clearly identifying or non-encryptable information is present, the targeted removal of these sensitive features is initiated in a subsequent work step. In this activity, the selective removal of data that could allow conclusions about individuals occurs, ensuring irreversible protection.

Following this string of measures, which is characterized either actively by the removal of individual information or proceeds directly without this intermediate step, both processing paths converge in a merging mechanism. This synchronization point ensures that in any case, a consolidated data set is available as a common basis for the next processing step. On this basis, further work is carried out in a central anonymization activity using predefined techniques. The previously determined or cleaned data is processed using explicitly defined data associations. It is ensured that the transformation not only leads to a change in the data but also to the permanent obfuscation of personal characteristics. The resulting anonymized database is transferred into a separate data object as a meaningful result.

With the transition to the final event, the successful completion of the entire process is signaled. This end event confirms that all necessary measures, from the initial assessment to the targeted removal of sensitive information to the final anonymization, have been consistently implemented. This ensures that any subsequent identification of the original personal data is impossible and the data is available in a state that fully meets the requirements for data protection and data security.

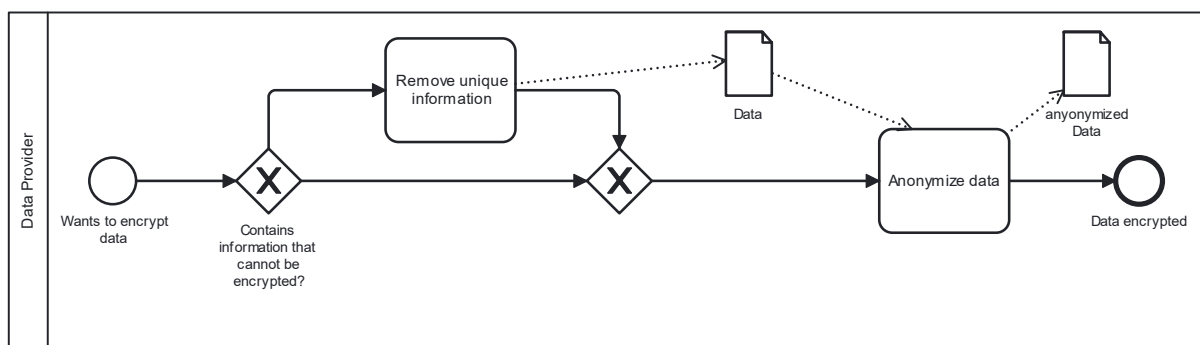


Figure 19. Anonymized Data Sharing.

Data Storage

Central Storage

Participating Actors: Data Provider, Data Trustee

Description: In this workflow, the central storage of data is realized within an infrastructure managed by the data trustee. The procedure is initiated by a start event that sets the storage of the data in a central database in motion. In the initial activity, the physical filing of the data takes place, with the information to be stored being simultaneously transferred via a data output association into a centrally referenced data store. This step forms the basis for further process execution.

During the storage process, there is a possibility that an error will occur. An error event (boundary event) linked to the storage activity monitors this step and catches occurring errors. In the event of an error, the regular workflow is interrupted, and an alternative path is initiated, culminating in a send task. Via this task, an error message is transmitted to the data provider, whereupon the process ends with a negative conclusion.

If, on the other hand, no disruption occurs during the storage process, the workflow continues by setting up access management in a subsequent activity. This step serves to configure security requirements and access rights that regulate access to the centrally stored data inventory. In parallel, an access notification is sent to the data provider via a defined message flow, ensuring that both actors are informed about the current status and the established access permissions. The successful completion of this activity leads in an end event to the confirmation that the data was successfully stored and managed in the central infrastructure.

Through this structured workflow, which includes both systematic data storage and integrated error handling, efficient and secure management of the data is ensured. The data trustee controls the entire process, while the data provider is informed early on about access or any errors via targeted communication channels.

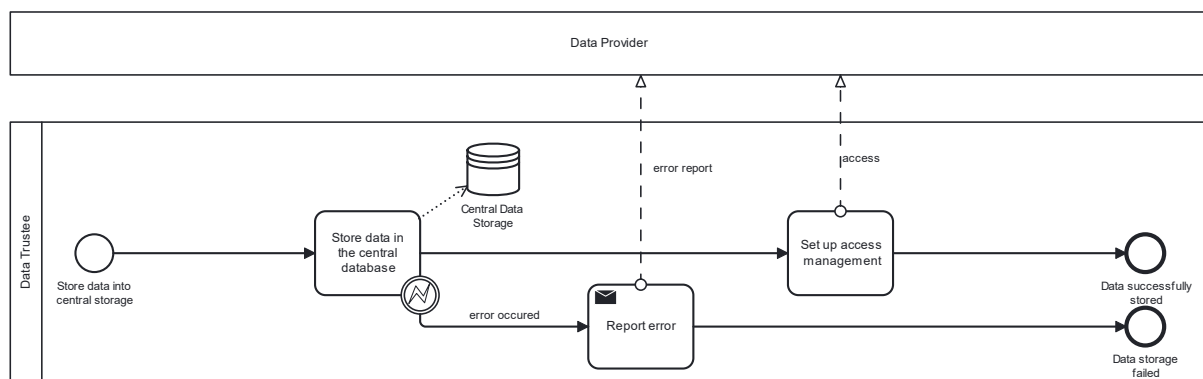


Figure 20. Central Storage.

Decentral Off-chain Storage

Participating Actors: Data Provider, Data Trustee, Distributed Systems

Description: In the illustrated workflow, the procedure begins with a triggering event signaling the order to record the data in a central storage system. Once the entity responsible for executing the process has received the necessary information, the first step is to calculate a cryptographic hash value of the handed-over data. Through this calculation, a unique fingerprint is generated, serving as the basis for further validations.

Subsequently, the information is actively transmitted to a decentralized storage network. This is done via a send activity in which a corresponding message, designated as a transaction, is sent to the external, distributed system. In parallel, the process sets another operation in motion that determines whether the previously calculated hash or, alternatively, a URL is to be used as a permanent reference for the stored data. This result forms the basis for the next essential activity.

The subsequent phase involves depositing a smart contract in which the determined reference link is saved. This smart contract is archived in an intended contract database, creating a tamper-proof, permanent record. Thus, the connection between the decentralized stored data sets and the blockchain, which acts as proof and reference, is bindingly documented.

After the successful storage of the smart contract, a query follows at an event-driven decision point as to whether the transaction has actually occurred in the decentralized system. At this gateway, either an incoming confirmation message signals that the data transmission was successfully completed, or a time-triggered event occurs if no confirmation arrives within a defined time window. In the first case, upon receipt of the confirmation message, the integrity of the decentralized stored data is checked in a subsequent activity. This review ensures that the data has remained unchanged and consistent during the transmission as well as in the decentralized system.

If the integrity check is successful, the final measure is taken, in which access management is set up. Within this activity, not only are the necessary access permissions configured, but a notification is also sent to the original data supplier. This message transmits the relevant information enabling the data provider to access the decentralized kept data. Thus, the process ends positively with a corresponding conclusion event that documents the successful completion of data storage.

If, however, triggered by the time-controlled event, no confirmation has been received from the decentralized system within the intended period, the alternative path of the decision point initiates the termination of the process with an error event. In this case, the operation is declared a failure, which requires a new check or other troubleshooting measures.

This workflow seamlessly integrates the exchange of messages among the participating actors. A transaction message is sent to the distributed system, based on which a confirmation later comes back. Additionally, after the successful setup of access management, a message regulating access is transmitted to the data supplier. Overall, this process ensures that the actual data is stored decentrally, while the blockchain, through the saved reference and the associated smart contract, is used as an auditable, immutable log.

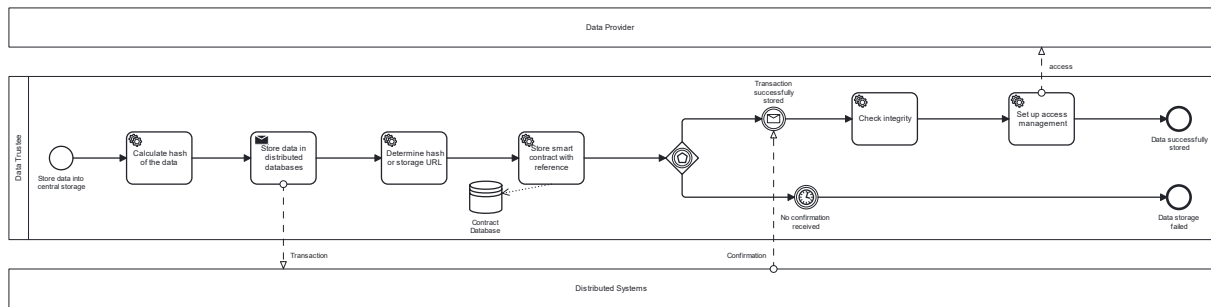


Figure 21. Decentral Off-chain Storage.

Decentral On-chain Storage

Participating Actors: Data Provider, Data Trustee, Blockchain Network

Beschreibung: The described workflow starts with an initiation event signaling that data (or its checksums) are to be deposited in a decentralized storage system. At this moment, the data is first hashed using an automated service, so that a unique checksum value is calculated serving as integrity-relevant proof. Subsequently, this information is incorporated into a transaction data record, which is transmitted to the blockchain network via a dedicated sending step. The transaction flows via a defined message channel from the executing system, which acts here as a trustee for data, directly to the network.

After the transaction is sent, the workflow reaches an event-driven branching element where two alternative reactions are awaited. On the one hand, the process waits for the receipt of a confirmation message from the blockchain network signaling that the transaction was successfully integrated into a block and thus anchored in the decentralized ledger. If this message is received, the workflow automatically initiates a step in which the anchoring of the process, i.e., the actual hardcoding of the transaction in a block, is documented by a corresponding activity. Building on this, a data integrity check follows, ensuring that the stored information is unaltered and tamper-proof. In the next step, an activity to set up access management is carried out, where the necessary permissions and access controls are configured using the underlying blockchain mechanisms or smart contracts. Via another message channel, information about the established access rights is transmitted to the data provider in the course of this step, allowing them to benefit from the defined access regulations in the future. Finally, the positive path of the workflow ends with an end event certifying the successful completion of data filing.

Alternatively, the workflow envisions that in the event of a failure to receive a corresponding confirmation message within a strictly defined period, a timer-activated event occurs. This signals that, despite the dispatch, no confirmation has been received from the blockchain network. As a result, the process leads immediately to an end event that documents the error in the storage process and indicates that the data could not be successfully introduced into the blockchain.

Overall, the process reflects close cooperation among the three participating actors: The trustee, who controls the internal processing steps from calculating the hash value to sending the transaction to checking data integrity and setting up access management; the blockchain network, which validates and confirms the transaction; and the data provider, who ultimately receives the access rights relevant to them via its own message channel. The use of message flows and event-driven branches ensures that the entire process is completed only upon valid and confirmed integration of the data into the blockchain, while a missing confirmation proof leads early to a defined abort.

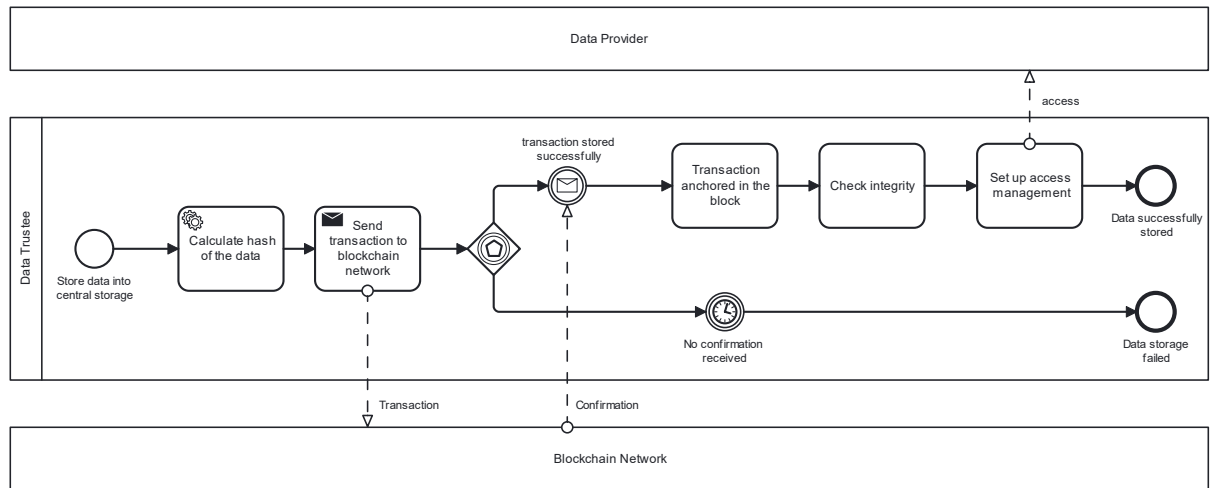


Figure 22. Decentral On-chain Storage.

Value-added Service

AI-as-a-Service

Participating Actors: Data Trustee, Data Consumer

Description: In the illustrated workflow, the requesting actor first expresses their desire to use the offered service. This initial impulse is initiated by a corresponding event that signals the need. Subsequently, the actor transmits their application using a defined interface implemented as a sending activity. This transmission occurs as a targeted message directed to the coordinating instance.

After sending the request, the process branches at an event-based decision point where two possible message receipts are queried as a reaction. On the one hand, it waits for a message signaling that access is denied. If this message arrives, the workflow leads via an intermediate event path directly to an end event documenting the denied access. In parallel, there is an alternative path waiting for positive feedback confirming successful access. If the requesting actor receives this confirmation, the process leads into an end event that concludes successful access.

While the requesting actor sends the application via the provided interface and reacts to the feedback, a separate coordinating unit, represented in the diagram as an independent instance, takes on the task of organizing the connection to the underlying AI platform. This unit provides either the necessary API or coordinates the connection to a suitable platform. In addition to these two central roles, another partner is mentioned who, however, is not explicitly integrated into the workflow steps or the message interactions in the present procedure.

The entire process is based on the exchange of messages triggered by specific message event definitions. This asynchronous communication allows the process to react flexibly to different responses, either a denial or a confirmation of access. This ensures that the requesting actor ultimately receives clear information about the status of their access. The process ends as soon as one of the two defined responses has been received, thereby conclusively documenting the usage process.

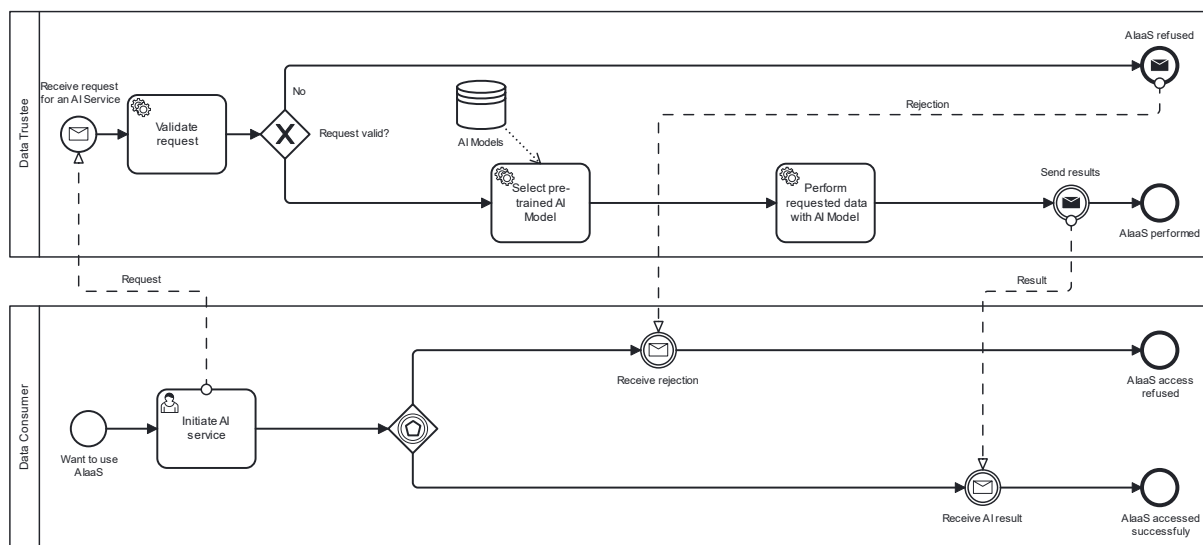


Figure 23. AI-as-a-Service.

AI Training

Participating Actors: Data Trustee, Data Consumer

Description: In this workflow, an initiating actor seeking to use an AI model communicates their desire to perform analyses via a send activity to a central service that checks and processes the incoming data. First, the requesting partner triggers their need by sending a message containing the analysis order. This message is received in the system of a specialized trustee operating in a separate organizational unit. There, the further course starts with the receipt of the application in the form of a message event, after which initial investigation steps are carried out immediately.

After the receipt of the training materials, provided by a separate data supplier, the content is analyzed in a first processing step. The input also flows via a data connection into a central data store, which is used as a reference for quality assurance. A subsequent decision point checks whether the initial data analysis meets the required quality criteria. In the event of a negative result, a rejection event is triggered via an alternative path, immediately sending a message back to the requesting partner and terminating the operation as rejected in their system.

If the initial check meets the expected criteria, the trustee continues processing. Another activity for data preparation takes place, where the training materials are prepared for iterative model training. Following these preparatory measures, the procedure is transferred to another decision node that controls the further course. Here, the process branches into two interlocking sub-workflows: On the one hand, an iterative cycle is initiated in which the model is trained and optimized within the framework of several successive activities, while on the other hand, alternative processing steps are worked through that may integrate additional preparation measures. The merging of both paths occurs in another gateway, the exit of which culminates in a message send event. The validated analysis results are transmitted to the incoming partner via this communication.

On the requesting partner's side, a waiting state is activated after sending the analysis order. An event-based gateway guides them either to the processing of the incoming results or, in the case of a rejection, to a corresponding termination step. If this partner receives the results, it is evaluated in a decision point whether the delivered information meets the requirements. A positive finding leads to the final conclusion of the procedure, which is thus successfully finished. Should a discrepancy be determined, however, a feedback mechanism prompts the repetition of the analysis order so that the entire cycle re-enters the process.

Overall, the defined workflow, which encompasses the exchange of messages, the execution of data checks, and iterative optimization and validation cycles, enables controlled and transparent processing of the training materials. The clear separation of roles between the provider of the training data, the trustee responsible for quality assurance and data preparation, and the requesting partner, who derives their decision from the results, ensures that when all quality criteria are met, the optimized AI model is finally integrated into the target system or made available to the receiving partner for use.

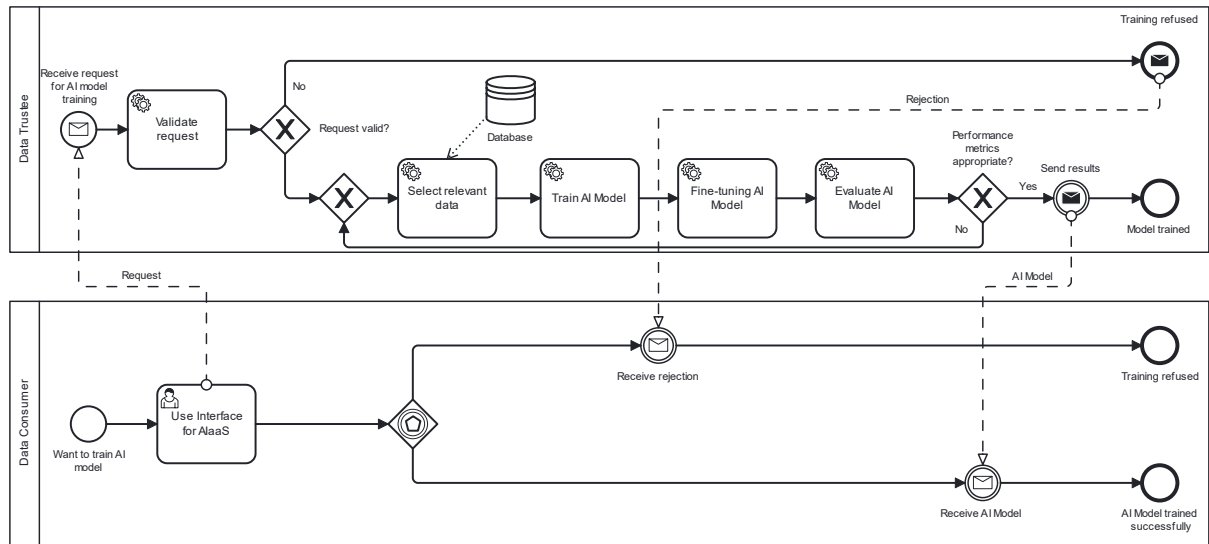


Figure 24. AI Training.

Data Analysis

Participating Actors: Data Trustee, Data Consumer

Description: In the presented scenario, two parties interact via clearly defined, coordinated sub-processes linked by message flows and decision points. The workflow is started on the data consumer's side, who expresses the desire to have data analyzed. For this purpose, a request is first triggered as part of a user task. This is transmitted as a message to the data trustee so that its process is set in motion via a message-based start event.

After receiving the request, the data trustee goes through a validation process. First, the request is checked using a service activity. A subsequent exclusive gateway evaluates whether the transmitted request meets the necessary criteria. In the case of a negative result branch, the data trustee initiates the path to rejection. The rejection is signaled in an end event that simultaneously generates a corresponding message. This rejection message is transmitted via a defined message flow to the data consumer, who receives it in their sub-process via an intermediate event, thus concluding the procedure.

If, conversely, the incoming request is considered valid, the process continues a multi-stage analysis workflow at the data trustee. In the first step, the relevant data sets are prepared as part of a preprocessing activity. Among other things, a referenced data store, the underlying database, is accessed. On this basis, statistical evaluations are carried out in a subsequent service activity. The insights gained here then flow into a further processing step in which a diagram visually summarizing the results is generated. As soon as the diagram is created, the transmission of the analysis results is initiated in an intermediate event. This message, summarizing the content of the analysis, specifically the generated diagram, is sent to the data consumer. The process at the data trustee then ends in an end event signaling the successful completion of data processing.

On the data consumer's side, the further course takes its way after sending the request via an event-based gateway. Here, the data consumer waits simultaneously for two possible responses: They receive either the analysis results in the form of the diagram or a rejection message documenting the negative decision path of the data trustee. In the event of a successful receipt of the diagram, a dedicated intermediate event takes effect, after which the data consumer evaluates the visual representation within a decision gateway. If the data visualization is correct and meets expectations, the corresponding path leads to an end event signaling successful completion. If it turns out, however, that the visualization does not meet expectations, a feedback loop is initiated that cyclically routes the workflow back to an earlier point, specifically to the initial decision node, so that the data consumer can make a request again.

The close coordination between the two participants is ensured through the use of defined message flows. Thus, the initial message flow links the start of the data analysis process at the data trustee with the request in the data consumer's system, while messages transmitted later in the form of rejections or analysis results control the further course in the data consumer's process. Through the use of exclusive and event-based gateways, the workflow is controlled in detail, from the validation decision to the sequential execution of data-driven activities to the iterative evaluation of the result. Overall, this workflow ensures that both data preparation and final evaluation are continuously adapted to the requester's needs and that a new analysis process can be triggered in the event of discrepancies.

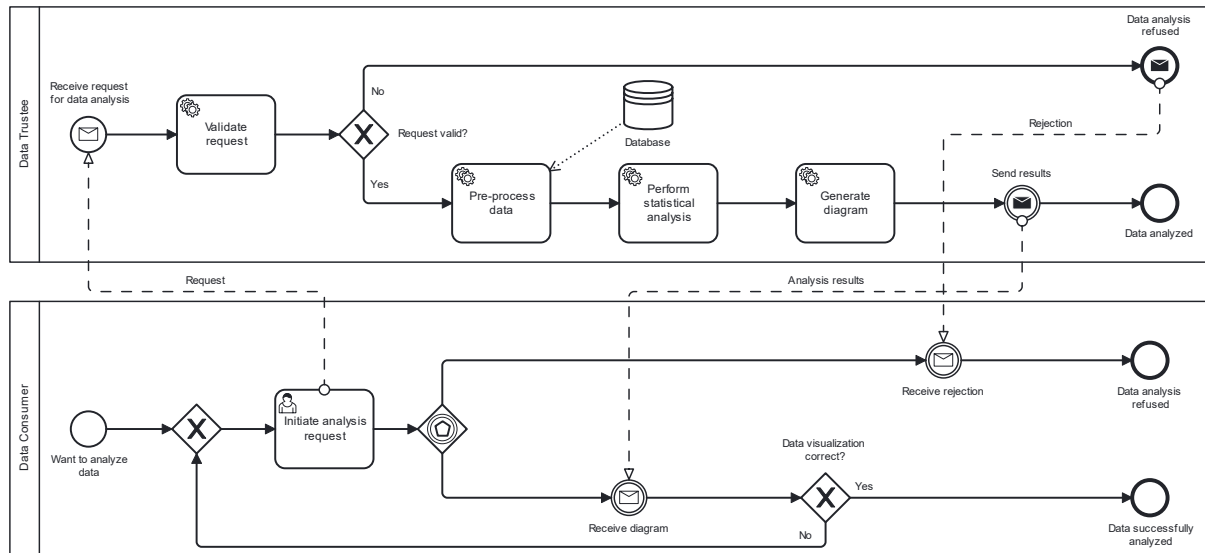


Figure 25. Data Analysis.

Quality Management

Participating Actors: Data Trustee, Data Consumer

Description: In the illustrated workflow, the data consumer initiates the entire process by signaling their need for a service to systematically monitor and improve data quality. Already in the first step, a corresponding task step occurs within their own workflows, the result of which triggers a message to request the service. This request is transmitted to the data trustee via defined message exchange flows.

As soon as the data trustee registers the request, specifically as a received message, its process starts with a validation of the incoming request. In the immediately following service activity, it is checked whether the transmitted requirements meet the specified criteria. A subsequent decision element determines whether the request can be classified as valid. If this is not the case, the procedure is terminated in an end event, which simultaneously generates a rejection message and sends it back to the data consumer.

If, conversely, the request is found to be valid, the next step involves defining the quality requirements. This activity serves to determine the framework conditions for the upcoming data analysis. Based on the defined requirements, a quality analysis process is then set in motion. First, a data profiling is carried out in a service activity, providing an initial detailed picture of the available data sets. The result of this profiling is forwarded to a decision element.

In the subsequent selection point, another gateway checks whether the results of the profiling are already sufficient. If the determined quality level is met, which is evaluated as “Quality sufficient?”, a corresponding path routes the result directly to the next coordination step. Should it be established, however, that initial profiling findings do not meet requirements, an additional service activity for validating data quality is executed. Here, too, a decision element is used, determining under the criterion “Quality OK?” whether the subsequent results meet standards. If the quality level is confirmed in this check step, or if it is reached after another iteration of profiling and validation, the process flows into a jointly converging decision instance.

This final merging occurs in a central gateway where all positive paths converge. Following this, the final result is triggered as a message in a special intermediate activity, which is then sent to the data consumer via a dedicated message channel. From the data trustee's side, the workflow ends in an end event that signals the state “Data available”.

In parallel, the data consumer follows its own workflow: After the initial request, they go through a work step in which they request the service. The process then reaches an event-based decision element that waits to see whether a message with the validated data or a rejection message is received. If they receive positive feedback, their process ends in a defined state that signals the successful completion of the service. Otherwise, upon receiving confirmation of rejection, their workflow concludes with a corresponding negative end event.

The structure of the entire process is characterized by the fact that it follows a strict review and validation procedure on the one hand, and enables continuous improvement of data quality through iterative feedback loops on the other. The close coordination between the two participants, via defined message exchange flows like “Request”, “Rejection”, and “Data”, ensures that it is clearly communicated at all times whether quality requirements are met or whether further measures are necessary. Overall, this workflow guarantees a transparent and dynamic process for ensuring and optimizing data quality.

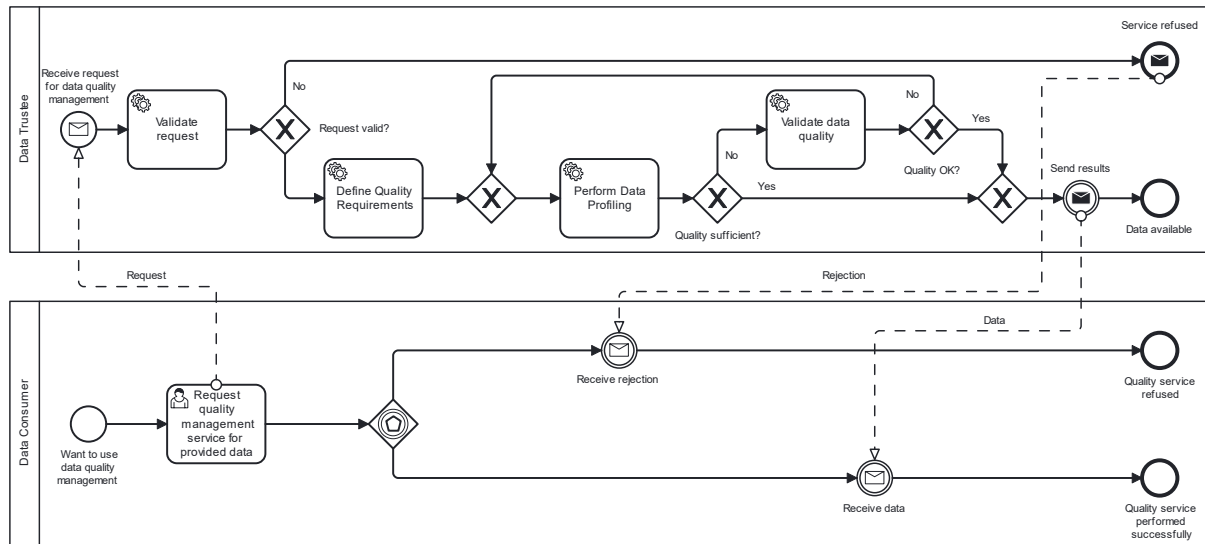


Figure 26. Quality Management.